



---

# Hardware-Aware FP4 FlashAttention-4 Backward

Adaptive low-precision gradients and unified QKV projection integration

## REPORT FOCUS

Adaptive FP4 Q/K, FP8 probability-gradient tensor work, BF16 dQ publication, and native NVFP4 projection epilogues.

**Robert Hu**

August 2026

---

## Abstract

This report documents a low-precision FlashAttention-4 backward for NVIDIA Blackwell, derived from the optimized ThunderKittens BF16 V382 schedule. The retained route is deliberately mixed precision: Q and K are adaptive E2M1 (FP4), score reconstruction uses three full-rank NVFP4 tensor instructions,  $dP$  and  $dV$  use E4M3 operands,  $dS$  is produced directly in E4M3, and  $dQ/dK$  use mixed E4M3×E2M1 tensor instructions. Tensor accumulators remain FP32. The best deployable endpoint reduces cross-owner  $dQ$  in BF16 and returns it directly to an adjacent projection-backward consumer.

Across causal B1,  $D_{QK} = 192$ ,  $D_V = 128$  shapes from S4096 to S16384, the direct-BF16 endpoint is 11.3–16.4% faster than the copied BF16 control. Calibrated cosine is 0.9914–0.9916 for  $dQ/dK$  and 0.99935–0.99938 for  $dV$ . Robust per-head Q/K scaling keeps sparse-outlier  $dQ/dK$  cosine near 0.962 instead of approximately 0.75 for the earlier global-amax rule.

The report also separates two often-confused operations. *Projection* means the learned linear maps  $[Q \mid K \mid V] = XW_{QKV}^T$ ;  $QK^T$  is the attention score contraction. A persistent NVFP4 projection now emits forward-native Q/K, transposed MXFP4 V, all backward Q/K views, and optional BF16 Q/K/V from the same output fragments. The compact Q/K payload is allocated once and shared by forward and backward. Relative to three BF16 GEMMs plus Q/K packing, unified publication is 1.41–1.68× faster with BF16 stores and 1.60–1.93× faster when they are suppressed on the two measured shapes. Pair-native full-head RoPE is now applied in the projection epilogue before all BF16 and FP4 Q/K publication. On a five-shape RoPE-aware boundary that keeps the FP4 FA4 forward in every low-precision row, the selected route is 1.096–1.469× faster than BF16 when projection weight gradients are charged and 1.118–1.720× faster for activation gradients. Useful algorithmic MFU rises on every row; at S4096/H24/K3072 it increases from 26.94% to 32.38% for the full training boundary. An unsigned stacked NVFP4 QKV-gradient handoff raises the activation-only result from 1.257× to 1.266×, whereas a signaled two-lane hierarchy is slower. A fused upstream dO producer also replaces separate row, column, and softmax-delta passes, reducing setup from 0.132352 to 0.068416 ms at S8192/H8. A compact delayed-scale NVFP4 dQ projection reaches approximately 0.991 cosine and improves the composed H24 chain by 3.7%, although it loses 1.9% at H8 and still follows a private BF16 dQ reduction. Pure-FP4 NVFP4/MXFP4 combinations do not displace the hybrid: the accurate NVFP4-dS rung is slower than BF16 and the combined NV/NV diagnostic is non-finite. The report states the represented equations, ownership and TMEM lifetimes, accuracy boundary, rejected alternatives, and the remaining cross-kernel optimization ceiling. In a three-seed, 500-step teacher/student proxy the mixed route retains 93.2% of BF16’s absolute validation-loss improvement on average, but its raw native loss remains higher because the FP4 forward contributes a measured quantization floor. This is block-level optimization evidence, not a stock-Llama perplexity or language-model convergence result.

## 1 Executive Summary

### 1.1 Terminology: projection is not the score product

Throughout this report, *projection* is reserved for the learned linear layers

$$Q = XW_Q^T, \quad K = XW_K^T, \quad V = XW_V^T. \quad (1)$$

The attention operation

$$S = \alpha QK^T, \quad \alpha = d^{-1/2}, \quad (2)$$

is called the *score contraction* or *score reconstruction*. This distinction matters because the latest speedup crosses the boundary between the learned QKV projection and attention: it emits the representations needed by forward and backward while Q, K, and V are still in projection-epilogue registers.

## 1.2 What is retained

The stable accuracy-oriented route is mode `kCta2DenseLowFp4Fp8DpPvReuseP`. Its precision contract is summarized in Table 1. The phrase *FP8 dP/dV* describes the matrix operands; it does *not* mean that the output gradients or TMEM accumulators are FP8.

**Table 1:** Retained backward precision contract. Represented denotes the decoded value seen by the tensor instruction.

| Operation               | A operand       | B operand                  | Accumulation / output                             |
|-------------------------|-----------------|----------------------------|---------------------------------------------------|
| score $QK^T$            | adaptive E2M1 Q | adaptive E2M1 K            | NVFP4 block scales, three K64 commands, FP32 TMEM |
| $dP = GV^T$             | E4M3 $G$        | E4M3 $V$                   | FP32 TMEM; represented at a fixed $2^4$ scale     |
| $dV = P^T G$            | E4M3 $P$        | E4M3 $G$                   | FP32 TMEM, then BF16 output                       |
| $dS = P \odot (dP - r)$ | E4M3 $P$        | FP32 $dP$ , FP16 centering | direct E4M3 production at scale $2^{12}$          |
| $dQ = dSK$              | E4M3 $dS$       | E2M1 $K$                   | mixed F8F6F4, FP32 TMEM, BF16 global reduction    |
| $dK = dS^T Q$           | E4M3 $dS$       | E2M1 $Q$                   | mixed F8F6F4, FP32 TMEM, one BF16 store           |

The route retains the complete  $D_{QK} = 192$  score rank. It does not drop the third K64 score command. Q/K use one robust adaptive multiplier per batch/head, shared by Q and K, so score reconstruction needs only the existing scale page. The complete BF16  $dS$  tile is never materialized: four FP32 N32 fragments are centered, multiplied by the already-quantized P representation, and converted directly to their final E4M3 consumer layouts.

## 1.3 Measured status

The current result has six distinct boundaries:

1. **Standalone attention backward.** Native BF16  $dQ$  publication is 11.3–16.4% faster than the copied BF16 V382 control on the measured S4096–S16384 matrix.
2. **Adaptive mixed-dP frontier.** A lifetime-safe specialization is 2.3–2.7% faster than adaptive FP8 dP/dV at H24, but 1.3–1.7% slower at H8 and reduces calibrated dQ/dK cosine to approximately 0.983.
3. **QKV projection production.** A persistent NVFP4 learned projection emits one shared forward/backward Q/K representation, transposed MXFP4 V, and optional BF16 Q/K/V. It avoids separate HBM rereads and fake-quantization launches. The legacy Q/K entry point retains its K-depth dispatcher; the unified entry point fuses all publication. Pair-native full-head RoPE is applied to Q/K register fragments before every BF16 and FP4 publication.
4. **Upstream dO production.** One fused kernel publishes row and column MXFP4 dO plus the exact softmax delta. It is bit-exact with the split quantizers and nearly halves setup time at S8192/H8.

5. **Projection backward.** Fully reduced BF16  $dQ, dK, dV$  can be written to one interleaved per-head buffer and consumed without three standalone tensors or concatenation. Per-head readiness inside the projection K loop plus two alternating TMEM accumulators now gives a small H8 win; wider heads shape-dispatch to BF16 dQ plus cuBLAS. A separate delayed-scale NVFP4 dQ handoff improves the H24 composed chain by 3.7% at approximately 0.991 dX cosine, but loses 1.9% at H8.
6. **RoPE-aware composed boundary.** Across S4096–S16384 and H24–H64, the upstream-native delayed-scale route is  $1.198\text{--}1.617\times$  faster than BF16 for activation gradients and  $1.166\text{--}1.424\times$  when projection weight gradients are also charged. Fusing inverse RoPE into delayed-scale NVFP4 dgrad packing is bit-exact. The no-publication and BF16-republication forms save 1.70% and 2.12% of that boundary component. At S4096/H24/K4096 this becomes a 0.46% activation-scope and 0.60% full-training improvement over the separate inverse-transform route.
7. **Stacked QKV-gradient and MFU boundary.** The normal unsigaled dQ endpoint can inverse-rotate, quantize, and project stacked dQ/dK/dV in one NVFP4 path. At S4096/H24/K3072 it reaches 1.566048 ms versus 1.983296 ms for BF16 ( $1.266\times$ ), while the signaled two-lane hierarchy is slower. Across five FP4-forward shapes, full-training speedup is  $1.096\text{--}1.469\times$  and activation speedup is  $1.118\text{--}1.720\times$ . A three-seed 500-step block-level proxy retains 93.2% of BF16’s absolute loss improvement on average, with a separately reported FP4 forward floor.

## 1.4 Outcome of the six integration experiments

**Table 2:** Keep/reject result for the six forward-style integration avenues.

| No. | Experiment                             | Outcome                                                                                                                                                                                                                                               |
|-----|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Unified learned QKV projection         | Retain: native Q/K, pair-native fused RoPE, MXFP4 V, and optional BF16 stores from one NVFP4 GEMM.                                                                                                                                                    |
| 2   | One Q/K representation                 | Retain: forward typed tensors and backward byte tensors alias one compact allocation.                                                                                                                                                                 |
| 3   | Represented P replay                   | Retain local E4M3 reuse; reject a 272 MiB forward P cache and the unstable MXFP4 P-to-dS candidate.                                                                                                                                                   |
| 4   | Upstream low-precision dO              | Retain fused row/column/delta publication; $1.935\times$ faster than the split setup.                                                                                                                                                                 |
| 5   | Direct projection-gradient consumption | Retain head-pipelined BF16 consumption at H8; retain the interleaved/vendor-GEMM fallback elsewhere. Retain compact NVFP4 as an explicit H24 experiment and the unsigaled stacked QKV NVFP4 path for activation dgrad; reject the signaled hierarchy. |
| 6   | Storage and layout reclamation         | Retain projection scratch aliasing and BF16 suppression; reject compact 96-byte mixed-dP transport because it does not retire on SM100.                                                                                                               |

The latest profile is not tensor-compute bound. At S8192/H8 the dense kernel has 21.0% tensor-pipe activity, 25.6% issue activity, 128 registers, no spills, and 231,484 bytes of shared memory. The largest sampled dependency is the wait for score issue/production. The 512-column TMEM

allocation is full, so a forward-like second complete score accumulator cannot be added without changing ownership or draining a persistent gradient accumulator.

## 1.5 Status labels

To avoid turning an experimental precision ladder into a production claim, this report uses five labels:

**Table 3:** Status of the low-precision routes discussed in this report.

| Label          | Route                                       | Meaning                                                                                                                                                                |
|----------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Retained       | adaptive FP4 Q/K + E4M3 $dP/dV/dS$ operands | stable zero-spill winner and public adaptive API                                                                                                                       |
| Frontier       | fixed-scale mixed E4M3/MXFP4 $dP$           | 3.3–5.3% faster than the retained route but lower $dQ/dK$ cosine and not the accuracy-oriented default                                                                 |
| Shape frontier | adaptive Q/K + mixed E4M3/MXFP4 $dP$        | lifetime-safe and faster at H24, but slower at H8 and approximately 0.983 calibrated $dQ/dK$ cosine                                                                    |
| Integration    | compact delayed-scale NVFP4 $dQ$ projection | approximately 0.991 $dX$ cosine; 3.7% H24 win and 1.9% H8 regression while private BF16 reduction scratch remains                                                      |
| Integration    | stacked delayed-scale NVFP4 QKV projection  | retained for activation dgrad with the normal unsigaled $dQ$ endpoint; the two-lane hierarchy is accurate but slower, and training keeps BF16 QKV for weight gradients |
| Experimental   | pure MXFP4 $dP, dV, dS, dQ, dK$ ladder      | represented-operand correctness established; not the selected long-sequence quality/performance point                                                                  |

The report is therefore a snapshot of the active implementation, not a claim that every backward arithmetic operation is FP4.

## 2 Backward Equations and Represented Formats

### 2.1 Exact attention backward

For one batch/head, let

$$Z = \alpha QK^T, \quad P = \text{softmax}(Z + M), \quad O = PV, \quad (3)$$

where  $M_{ij} = 0$  for legal entries and  $-\infty$  for masked entries. Given the upstream gradient  $G = \partial\mathcal{L}/\partial O$ , exact backward is

$$dV = P^\top G, \quad (4)$$

$$dP = GV^\top, \quad (5)$$

$$r_i = \sum_j P_{ij} dP_{ij} = \langle O_i, G_i \rangle, \quad (6)$$

$$dS = P \odot (dP - r\mathbf{1}^\top), \quad (7)$$

$$dQ = \alpha dSK, \quad (8)$$

$$dK = \alpha dS^\top Q. \quad (9)$$

The identity in Equation 6 lets the kernel compute the row correction from saved  $O$  and  $G$  before score reconstruction. Causal masking zeros all  $j > i$  entries in  $P$  and therefore in  $dS$ .

As in FlashAttention, the implementation tiles the sequence dimensions and recomputes scores and probabilities from  $Q$ ,  $K$ , and the saved log-sum-exp  $L_i$ ; no  $S \times S$  matrix is materialized in HBM [1, 8].

## 2.2 Quantizer notation

Let  $Q_F(x)$  be round-to-nearest, saturating conversion to format  $F$ . For a positive encode multiplier  $m$ , define

$$\bar{x}_F = Q_F(mx), \quad \hat{x}_F = m^{-1}\bar{x}_F. \quad (10)$$

An overbar denotes the stored code interpreted as its small floating value; a hat denotes the decoded represented value. E2M1 has the nonnegative magnitudes

$$\mathcal{F}_{\text{E2M1}}^+ = \left\{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3, 4, 6\right\}, \quad (11)$$

plus sign. E4M3 is used both as a payload format and as the scale format for NVFP4. MXFP4 instead pairs E2M1 blocks of 32 values with E8M0 power-of-two scales [4, 7].

## 2.3 Robust per-head FP4 Q/K

For batch/head index  $h$ , define

$$a_h = \max(\|Q_h\|_\infty, \|K_h\|_\infty), \quad (12)$$

$$\rho_h = \max\left(\sqrt{\frac{\|Q_h\|_F^2}{Sd}}, \sqrt{\frac{\|K_h\|_F^2}{Sd}}\right). \quad (13)$$

The retained clipping magnitude is

$$c_h = \max(0.325 a_h, 2.75 \rho_h). \quad (14)$$

With implementation bounds  $m_{\min}$  and  $m_{\max}$ , the target multiplier is

$$m_h^* = \begin{cases} \text{clip}(6/c_h, m_{\min}, m_{\max}), & c_h > 0, \\ m_{\max}, & c_h = 0. \end{cases} \quad (15)$$

The tensor instruction consumes an E4M3 dequantizer. The producer therefore snaps the reciprocal first,

$$\delta_h = Q_{\text{E4M3}}\left((m_h^*)^{-1}\right), \quad m_h = \delta_h^{-1}, \quad (16)$$

then packs

$$\bar{Q}_h = Q_{\text{E2M1}}(m_h Q_h), \quad \bar{K}_h = Q_{\text{E2M1}}(m_h K_h), \quad (17)$$

$$\hat{Q}_h = \delta_h \bar{Q}_h, \quad \hat{K}_h = \delta_h \bar{K}_h. \quad (18)$$

Q and K deliberately share  $m_h$ . Separate multipliers would require more score scale state and a different instruction-facing layout.

## 2.4 Full-rank FP4 score reconstruction

$D_{QK} = 192$  is split into three K64 chunks. The represented base-two score is

$$\tilde{Z}_2 = \alpha \log_2(e) \sum_{c=0}^2 \text{MMA}_{\text{NV4}}\left(\bar{Q}_{h,c}, \bar{K}_{h,c}^\top; \delta_h, \delta_h\right), \quad (19)$$

where the semicolon denotes the E4M3 scale page consumed by the block-scaled instruction. All three chunks are retained. The two-command D128 approximation is disabled.

Let  $L_i$  be the log-sum-exp saved by the forward that produced O. The backward probability used by the represented operator is

$$\tilde{P}_{ij} = 2^{\tilde{Z}_{2,ij} - L_i \log_2(e)} \mathbf{1}_{\{j \leq i\}}. \quad (20)$$

Because  $L_i$  belongs to the forward operator while the score is reconstructed from represented Q/K, Equation 20 is not, in general, the exact softmax of  $\hat{Q}\hat{K}^\top$ . This is the ordinary saved-LSE backward contract, now evaluated with low-precision Q/K.

## 2.5 Fixed-power E4M3 path

The retained producer uses exact powers of two so scale work becomes exponent adjustment. Define

$$\bar{G}_4 = Q_{\text{E4M3}}(2^2 G), \quad \bar{V}_4 = Q_{\text{E4M3}}(2^2 V), \quad \bar{P}_8 = Q_{\text{E4M3}}(2^8 \tilde{P}). \quad (21)$$

The subscripts 4 and 8 indicate the power-of-two multiplier, not bit width. The  $dP$  tensor instruction produces

$$D_P = \text{MMA}_{\text{E4M3}}(\bar{G}_4, \bar{V}_4^\top) \approx 2^4 dP. \quad (22)$$

The mandatory statistics pass publishes the matching row term directly:

$$R_i = 2^4 \langle O_i, G_i \rangle. \quad (23)$$

Thus  $D_P - R\mathbf{1}^\top$  can be centered without first dequantizing the whole  $dP$  tile.

For  $dV$ , the same  $2^2$  E4M3  $G$  operand is reused:

$$\widetilde{dV} = 2^{-10} \text{MMA}_{\text{E4M3}}(\bar{P}_8^\top, \bar{G}_4). \quad (24)$$

The  $2^{-10}$  correction is exactly  $2^{-(8+2)}$ . This is why the retained path does not need a second  $2^8 G$  representation.

## 2.6 Fused E4M3 dS and mixed FP8-by-FP4 gradients

E4M3 P is unpacked to FP16 pairs, while scaled  $dP$  and  $R$  are centered in FP16. Abstracting that local rounding as  $\text{fl}_{16}$ , the published dS code is

$$\overline{dS}_{12} = Q_{\text{E4M3}}(\text{fl}_{16}(\bar{P}_8) \odot \text{fl}_{16}(D_P - R\mathbf{1}^\top)) \approx Q_{\text{E4M3}}(2^{12}dS). \quad (25)$$

The factors  $2^8$  and  $2^4$  multiply to the required fixed dS scale  $s_{dS} = 2^{12} = 4096$ . Conversion is performed quarter by quarter from the FP32 producer fragment into both instruction-facing orientations. No complete BF16 dS tile exists.

Finally,

$$\widetilde{dQ} = \frac{\alpha}{2^{12}m_h} \text{MMA}_{\text{E4M3} \times \text{E2M1}}(\overline{dS}_{12}, \bar{K}_h), \quad (26)$$

$$\widetilde{dK} = \frac{\alpha}{2^{12}m_h} \text{MMA}_{\text{E4M3} \times \text{E2M1}}(\overline{dS}_{12}^\top, \bar{Q}_h). \quad (27)$$

The scalar corrections are fused into the existing dQ publication and dK epilogue. FP32 tensor accumulation is preserved before BF16 publication.

## 2.7 Seven-word scale record

Each batch/head has one FP32 record:

$$\mathcal{M}_h = \left[ m_h, m_h, \frac{\alpha}{2^{12}m_h}, \frac{\alpha}{2^{12}m_h}, \frac{\alpha \log_2 e}{m_h^2}, u_h, \text{bits}(\delta_h, \delta_h, \delta_h, \delta_h) \right]. \quad (28)$$

The first two words drive Q/K packing; words 2 and 3 are the live dQ/dK corrections; word 5 is producer completion scratch; word 6 is the repeated E4M3 scale-page byte pattern. Word 4 keeps the explicit scalar score factor in the interface contract, while the current hot score path applies the equivalent  $\delta_h$  scale page plus its ordinary  $\alpha \log_2 e$  factor.

# 3 Blackwell Ownership and Memory Architecture

## 3.1 The copied V382 schedule is the control

The low-precision tree is a namespace-isolated copy of the optimized ThunderKittens BF16 V382 backward [3]. The BF16 source is kept as the invariant control: low precision changes representations and selected producer/consumer handoffs inside the same causal scheduler rather than replacing it with a small standalone matrix benchmark.

The retained specialization uses one two-CTA cluster. Each CTA has 512 threads divided into loader, tensor-issue, statistics, compute, and reducer roles. A K/V owner keeps its  $dK$  and  $dV$  accumulators resident while it visits the legal query tiles. The two CTAs exchange the fragments needed to form full dQ and dK operands. This owner topology is the reason  $dK$  and  $dV$  are single-writer outputs while dQ is a cross-owner reduction.

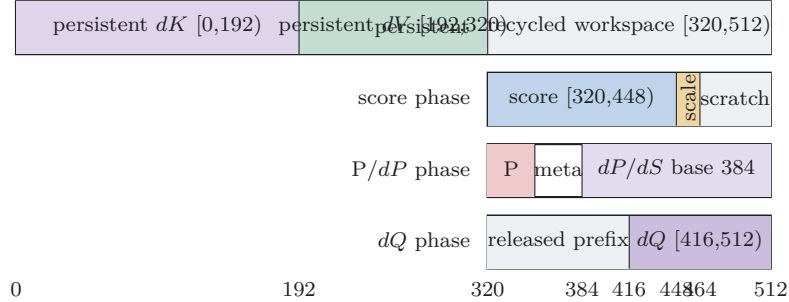


**Table 4:** Inherited schedule versus the active low-precision changes.

| Component     | Inherited from BF16 V382                                 | Active low-precision change                                  |
|---------------|----------------------------------------------------------|--------------------------------------------------------------|
| Ownership     | two-CTA K/V owner; persistent causal query loop          | unchanged                                                    |
| Tile geometry | $M128 \times N128$ , $D_{QK} = 192$ , $D_V = 128$        | unchanged                                                    |
| Score         | three K64 reductions into FP32 TMEM                      | NVFP4 Q/K operands and one adaptive scale page               |
| P and dS      | recompute P from saved LSE; produce both dS orientations | P is E4M3 and reused; dS is emitted directly as E4M3         |
| $dP, dV$      | BF16 tensor operands, FP32 accumulation                  | E4M3 tensor operands with exact power-of-two corrections     |
| $dQ, dK$      | BF16 tensor operands, FP32 accumulation                  | mixed E4M3 $\times$ E2M1 tensor instructions                 |
| Publication   | cross-owner dQ reduction; single stores for dK/dV        | dQ reduction payload becomes BF16; dK/dV ownership unchanged |

### 3.2 TMEM is full and heavily aliased

The active kernel uses all 512 logical TMEM columns. Long-lived dK and dV occupy columns 0–319. Score, P, dP, dS, scale pages, and dQ time-share columns 320–511. Figure 1 shows logical bases and the dominant lifetimes; the lower rows are temporal overlays, not simultaneous independent allocations.



**Figure 1:** Retained logical TMEM map. dK spans a 128-column main tile and a 64-column tail; dV spans 128 columns. The score uses [320,448), its scale page starts at 448, and the compact dQ accumulator uses [416,512). P, dP, and dS reuse retired score/dQ addresses according to phase.

The most important consequence is negative: another complete  $M128 \times N128$  FP32 score bank needs 128 columns, but no such bank exists. Splitting a score or dQ issue can expose an earlier prefix, yet barriers cannot manufacture storage. Experiments that drain half of persistent dV to buy columns lose more to scratch traffic than they gain from score lookahead.

### 3.3 The alias boundary defines legal issue order

The transient region follows the ownership chain

$$\text{score} \rightarrow \text{P publication} \rightarrow dP/dV \text{ consume} \rightarrow dS \rightarrow dQ \text{ consume} \rightarrow \text{next score.} \quad (29)$$

The next score cannot overwrite columns 416–447 until the current dQ consumer has captured that prefix. Conversely, delaying score issue until all dQ publication completes exposes unnecessary latency. The retained schedule uses issue markers: score and dQ consumers may enqueue their TMEM loads after the producer instruction has been issued, then let the native TMEM fence cover completion. Full completion barriers remain for final drain and lifetime proof.

### 3.4 Shared memory and register boundary

The profiled hot specialization uses 231,484 bytes of dynamic shared memory, 128 registers per thread, and no local stack or spills. Shared memory carries the TMA-fed Q/K/V/G views, Q/dS peer exchange, statistics, and instruction descriptors. The kernel is already at the practical register boundary: an additional score buffer is ruled out by TMEM capacity. The original adaptive mixed-dP attempt also aliased its fixed dP scale word with the live adaptive score scale page, causing repeatable tile-local corruption. Giving those values distinct lifetime-safe pages produces a stable 128-register, zero-spill specialization. Useful further pipelining still needs a different ownership topology or an upstream/downstream fusion, not another local barrier stage.

### 3.5 Only dQ is a global reduction

For an owner  $o$ , the kernel forms a partial

$$dQ^{(o)} = \alpha dS^{(o)} K^{(o)}, \quad dQ = \sum_o dQ^{(o)}. \quad (30)$$

The retained implementation publishes these partials with `cp.reduce.async.bulk.tensor` addition. It is not a scalar atomic loop. At S8192/H8 the FP32 version moved 830.5 MB into the reduction path, 16.5 times the final 50.3 MB FP32 dQ tensor. The retained BF16 reduction halves that payload. dK and dV stay in TMEM for their complete owner lifetime and are stored once, so there is no dK/dV atomic mechanism to replace.

## 4 Retained Adaptive FP4+FP8 Kernel

### 4.1 End-to-end dataflow

Algorithm 1 describes one owner iteration. It is a dependency description rather than a serialized instruction listing: independent loader, tensor, statistics, compute, and reducer roles overlap whenever the lifetime constraints of Figure 1 permit.

### 4.2 Quantization is fused where the value is born

The implementation follows one rule: conversion should happen in the producer that already owns the FP32 or BF16 fragment. In particular:

- $G$ ,  $V$ , row dot products, and LSE conversion share one mandatory linear preprocessing pass.
- $P$  is converted directly from the score fragment and stored in its final E4M3 TMEM orientation.

---

**Algorithm 1** Retained adaptive low-precision backward for one K/V owner

---

| Owner            | Action                                                                                                              |
|------------------|---------------------------------------------------------------------------------------------------------------------|
| 0 preprocessing  | Compute $r = \langle O, G \rangle$ and $L \log_2 e$ ; pack $2^2 G$ and $2^2 V$ to E4M3; overlap the BF16 dQ clear.  |
| 1 loader         | Bring compact adaptive Q/K, V, and $G$ views into their instruction-facing shared-memory stages.                    |
| 2 tensor issuer  | Issue three K64 NVFP4 score commands into TMEM [320,448), using the per-head E4M3 scale page.                       |
| 3 compute warps  | Read N32 score fragments, apply saved LSE and causal mask, and publish $Q_{E4M3}(2^8 \tilde{P})$ directly to TMEM.  |
| 4 tensor issuer  | Issue E4M3 $dP = GV^\top$ and $dV = P^\top G$ commands into their FP32 accumulators.                                |
| 5 compute warps  | Center scaled dP by $2^4 r$ , multiply by the represented P, and emit both E4M3 dS orientations at scale $2^{12}$ . |
| 6 tensor issuer  | Issue mixed E4M3×E2M1 dK and dQ commands; retain dK/dV and publish dQ issue markers.                                |
| 7 reducers       | Drain dQ in x32 fragments, apply $\alpha/(2^{12} m_h)$ , and perform BF16 TMA reductions to global memory.          |
| 8 owner epilogue | Apply the dK and dV corrections and store each once; release the persistent owner state.                            |

---

- dS is converted directly from four N32 producer fragments. The code simultaneously forms local-dQ, peer-dQ, and dK-facing layouts.
- Q/K projection integration, described in Section 5, creates the four E2M1 layouts from projection-epilogue registers rather than rereading BF16 Q/K.

This is materially different from fake quantization, in which a BF16 tensor is written to HBM and a separate kernel rereads, quantizes, and transposes it. The fallback packers remain useful for validation and shallow projection depths, but they are not the intended integrated dataflow.

### 4.3 P reuse joins dV and dS

The E4M3 P code consumed by dV is also the value used in Equation 25. Reusing the exact bits has two benefits. It removes a second BF16 P representation, and it ensures that the softmax derivative is taken with respect to the same represented P used by the dV tensor product. The fixed powers satisfy

$$(2^8 P) [2^4 (dP - r)] = 2^{12} dS, \quad (31)$$

so the producer needs packed half arithmetic and one native E4M3 conversion, not general scale multiplies.

The row term is published as FP16 after being multiplied by  $2^4$ . This cuts the shared loads and registers of the eight native-x32 consumers. The final two words of the first P fragment are pre-unpacked into otherwise-unused registers, shortening the dependent dS loop without extending its live range.

#### 4.4 Score and dQ use issue-before-completion handoffs

Score consumers do not first spin on a full completion event. Once all three NVFP4 commands are queued, the tensor role publishes `score_issued`; the compute warps enqueue synchronous TMEM reads and rely on `tensor_load_wait` for actual data readiness. The ordinary `score_done` event stays live for statistics and final drain.

dQ uses the same pattern. The tensor role publishes `dq_issued` after the mixed dQ command is queued. Reducers begin their progressive prefix loads, allowing columns 416–447 to retire earlier for the following score. This reduces exposed producer-to-consumer latency without allocating another TMEM accumulator or weakening the final lifetime proof.

#### 4.5 BF16 dQ is the projection-ready endpoint

The default public API preserves an FP32 dQ result: owners reduce into BF16, then a linear epilogue widens the completed tensor. When the next learned linear backward accepts BF16, `return_bf16_dq=True` omits that pass and the FP32 allocation. The represented operation is unchanged; the remaining difference is the order of nondeterministic owner reductions.

The direct-BF16 endpoint is the retained deployable boundary. Suppressing dQ publication entirely measured a lower floor, but a correct consumer cannot use a partial tile:

$$dQ_{\text{complete}}[I, :] = \sum_o dQ^{(o)}[I, :]. \quad (32)$$

Under K/V ownership, the contributing owners are unclustered. Removing the global BF16 reduction requires a q-centric schedule or a larger fused projection/reduction topology.

#### 4.6 Current compile-time state

The result tables use:

```
TK_FA4_BWD_FP8DPDV_BF16_DQ_REDUCTION = 1,
TK_FA4_BWD_FP8DPDV_TWO_COMMAND_SCORE = 0,
TK_FA4_BWD_FP8DPDV_HALF_DQ_REDUCTION_CEILING = 0,
TK_FA4_BWD_FP8DPDV_SKIP_DQ_REDUCTION_CEILING = 0,
TK_FA4_BWD_FP8DPDV_SKIP_DQ_WIDEN_CEILING = 0.
```

The hot specialization remains at 128 registers with zero spill loads, zero spill stores, and the exact three-command score.

### 5 Learned Projection Integration

#### 5.1 What projection means

Flatten batch and sequence into  $M = BS$ , let the model hidden width be  $K$ , and concatenate the Q/K/V weights:

$$W_{QKV} = \begin{bmatrix} W_Q \\ W_K \\ W_V \end{bmatrix}, \quad Y_{QKV} = XW_{QKV}^T = [Q \mid K \mid V]. \quad (33)$$

This is a learned GEMM with reduction dimension  $K$ . It is separate from the attention score contraction in Equation 2, whose reduction dimension is the head dimension 192.

The previous integration ceiling assumed that an upstream projection could somehow emit the adaptive E2M1 operands. The current implementation makes that producer real: a persistent SM100 NVFP4 GEMM computes Equation 33 and publishes the forward-native Q/K E2M1 payloads, the backward Q/K views, and transposed MXFP4 V directly from its output register fragments. BF16 Q/K/V stores are optional rather than part of the low-precision contract. Q and K have per-head depth 192; V has depth 128. Packed weight rows are ordered as all Q heads, all K heads, then all V heads.

## 5.2 NVFP4 learned-projection equation

For a 16-value activation block  $b = (i, c)$ , containing one row and columns  $16c:16(c+1)$ , the standalone quantizer selects one global dequantizer  $g_X$  and one local E4M3 dequantizer  $s_{i,c}^X$ . In simplified notation,

$$g_X \approx \frac{\|X\|_\infty}{6 \cdot 448}, \quad (34)$$

$$s_{i,c}^X = Q_{\text{E4M3}} \left( \frac{\max_{16c \leq k < 16(c+1)} |X_{ik}|}{6g_X} \right), \quad (35)$$

$$X_{ik}^4 = Q_{\text{E2M1}} \left( \frac{X_{ik}}{g_X s_{i,c}^X} \right), \quad \hat{X}_{ik} = g_X s_{i,c}^X X_{ik}^4. \quad (36)$$

Activations retain this  $1 \times 16$  rule because each token row is an independent sample.

Learned weights instead use a transpose-consistent  $16 \times 16$  rule. For  $B_{r,c} = \{(j, k) : 16r \leq j < 16(r+1), 16c \leq k < 16(c+1)\}$ ,

$$s_{r,c}^W = Q_{\text{E4M3}} \left( \frac{\max_{(j,k) \in B_{r,c}} |W_{jk}|}{6g_W} \right), \quad (37)$$

$$W_{jk}^4 = Q_{\text{E2M1}} \left( \frac{W_{jk}}{g_W s_{r,c}^W} \right), \quad \hat{W}_{jk} = g_W s_{r,c}^W W_{jk}^4. \quad (38)$$

The same local scale is repeated into the sixteen hardware  $1 \times 16$  scale records covered by  $B_{r,c}$ . Consequently packing  $W^\top$  does not choose a second, incompatible set of scales:

$$\mathcal{Q}_{\text{NV4}}^{2D}(W^\top) = \Pi_\top \left( \mathcal{Q}_{\text{NV4}}^{2D}(W) \right), \quad (39)$$

where  $\Pi_\top$  transposes payload blocks and their scale page. This is a representation invariant for forward and dgrad, rather than a claim that the larger block minimizes one-shot quantization error.

The persistent projection computes

$$\hat{Y}_{ij} = \sum_{k=1}^K \hat{X}_{ik} \hat{W}_{jk} = g_X g_W \sum_b \text{MMA}_{\text{E2M1}} \left( X_{i,b}^4, W_{j,b}^4; s_{i,b}^X, s_{j,b}^W \right), \quad (40)$$

accumulates in FP32 TMEM and multiplies by the two global dequantizers. The implementation uses M256×N256 output tiles, K256 reduction steps, a two-CTA cluster, and a four-stage input ring. Each output fragment is rounded once to BF16 before its optional BF16 store and low-precision publication, so all published views share one rounding point.

### 5.3 Pair-native RoPE in the projection epilogue

The first composed forward/backward evaluation omitted rotary position embedding and therefore did not represent the Llama attention boundary. The unified QKV projection now optionally applies full-head RoPE to Q and K before either BF16 or low-precision publication. For  $D_{QK} = 192$ , write the usual split-half rotary coordinates of one unrotated head as

$$u = [a_0, \dots, a_{95}, b_0, \dots, b_{95}]^T, \quad \Pi u = [a_0, b_0, a_1, b_1, \dots, a_{95}, b_{95}]^T. \quad (41)$$

The Q and K projection-weight rows are converted once to the  $\Pi$  order. Because  $\Pi$  is orthogonal and is common to Q and K,

$$(\Pi q)^T (\Pi k) = q^T k, \quad (42)$$

so this storage choice does not alter the unrotated score operator. It makes each rotary plane local to one adjacent BF16 pair in the projection epilogue.

For token  $t$  and rotary plane  $p$ , define

$$R_{t,p} = \begin{bmatrix} c_{t,p} & -s_{t,p} \\ s_{t,p} & c_{t,p} \end{bmatrix}, \quad \tilde{u}_{t,p} = R_{t,p}(\Pi u_t)_p. \quad (43)$$

The epilogue applies Equation 43 to the rounded Q/K register fragment and then forms every consumer representation from that rotated fragment:

$$\tilde{Q}_R = \mathcal{R}_t(\Pi Q_R), \quad C_Q = Q_{E2M1}(m_h^Q \tilde{Q}_R), \quad (44)$$

$$\tilde{K}_R = \mathcal{R}_t(\Pi K_R), \quad C_K = Q_{E2M1}(m_h^K \tilde{K}_R). \quad (45)$$

Consequently the optional BF16 Q/K tensors, forward score operands, and all backward Q/K byte layouts represent the same rotated operator. V is not modified. The packed implementation evaluates both coordinates with one BF16x2 multiply and one BF16x2 fused multiply-add; this deliberately accepts one additional BF16 rounding at the already-BF16 publication boundary rather than expanding every live pair into FP32 registers.

Attention backward returns derivatives with respect to the rotated tensors. Before projection dgrad or weight-gradient GEMMs, each Q/K pair is mapped back to the pair-native projection output by

$$d(\Pi u_t)_p = R_{t,p}^T d\tilde{u}_{t,p}. \quad (46)$$

The reference implementation applies Equation 46 in-place to the Q and K fields of the interleaved BF16  $[dQ \mid dK \mid dV]$  tensor;  $dV$  is bit-preserved. A model may keep Q/K weights and their optimizer state in pair-native row order. Export to a split-half checkpoint requires only the inverse row permutation  $\Pi^T$ .

A projection-dgrad specialization now fuses the same inverse map into its delayed-scale NVFP4 operand producer. For a 16-value projection-input block  $b$ , it publishes

$$C_{t,b}^{\nabla,4} = Q_{E2M1} \left( \frac{\mathcal{R}_t^T [d\tilde{Q}_t \mid d\tilde{K}_t \mid dV_t]_b}{g_{\nabla} s_{t,b}^{\nabla}} \right), \quad (47)$$

where  $\mathcal{R}_t^T$  acts on Q/K pairs and is the identity on V. The transform occurs after the attention boundary's BF16 rounding and before local amax selection, so its E2M1 payload and scale page are bit-exact with the reference sequence “inverse RoPE, then NVFP4 quantize.” An optional store republishes the inverse-rotated BF16 tile for projection weight-gradient consumers. That store preserves the reference bits and is not a performance penalty in the measured kernel: its asynchronous BF16 publication overlaps the packing tail. Both forms are retained, with republication selected when projection weight-gradient consumers require the inverse-rotated BF16 tile.

## 5.4 One Q/K payload shared by forward and backward

For a BF16-rounded Q register slice  $Q_R$  and its head multiplier  $m_h^Q$ , the epilogue computes

$$C_Q = Q_{\text{E2M1}}(m_h^Q Q_R). \quad (48)$$

The same codes are rearranged into:

$$\mathcal{L}_{Q,\text{seq}}(C_Q) \in \text{uint8}^{B \times H \times 192 \times S}, \quad (49)$$

$$\mathcal{L}_{Q,\text{depth}}(C_Q) \in \text{uint8}^{B \times H \times S \times 96}. \quad (50)$$

For K it independently forms  $C_K = Q_{\text{E2M1}}(m_h^K K_R)$  and publishes

$$\mathcal{L}_{K,\text{aligned}}(C_K) \in \text{uint8}^{B \times S \times H \times 192}, \quad (51)$$

$$\mathcal{L}_{K,\text{depth}}(C_K) \in \text{uint8}^{B \times H \times S \times 96}. \quad (52)$$

The compact tensors  $\mathcal{L}_{Q,\text{depth}}$  and  $\mathcal{L}_{K,\text{depth}}$  are the storage allocation, not copies made for backward. Forward obtains typed E2M1 views of the same bytes while backward consumes uint8 views. The forward scale hierarchy is

$$s_{\text{local}}^Q = s_{\text{local}}^K = 448, \quad (53)$$

$$g_h^Q = \frac{1}{448 m_h^Q}, \quad g_h^K = \frac{1}{448 m_h^K}, \quad (54)$$

$$\hat{Q}_R = g_h^Q s_{\text{local}}^Q C_Q = \frac{C_Q}{m_h^Q}, \quad \hat{K}_R = g_h^K s_{\text{local}}^K C_K = \frac{C_K}{m_h^K}. \quad (55)$$

Thus forward and backward consume one represented Q/K operator. There is no second forward payload and no conversion between the two APIs.

Together with the optional BF16 Q/K store, one register slice can therefore emit the ordinary tensor and its consumer-specific E2M1 views. Aligned-U4 rows use eight E2M1 bytes followed by an eight-byte hole for every 16 logical values. Compact depth rows pack two E2M1 values per byte without those holes.

The four low-precision tensors are bit-exact against the standalone precomputed-scale packer. Fused and separate publication therefore have identical accuracy; their difference is only movement and scheduling.

## 5.5 Single-quantization pure-FP4 Q/K fanout

The pure-FP4 backward needs two additional compact Q/K operand views for its block-scaled dQ and dK instructions. A naive unified producer first formed the four adaptive forward/backward views above, then quantized the same Q/K register fragment a second time at the pure route's fixed multiplier. The aggressive specialization instead performs one fixed-scale quantization,

$$C_X^{(16)} = Q_{\text{E2M1}}(16X_R), \quad X \in \{Q, K\}, \quad (56)$$

and fans those codes directly into every consumer layout,

$$\mathcal{B}_{QK}^{\text{pure}} = \left\{ \Pi_\ell \left( C_{\chi(\ell)}^{(16)} \right) \right\}_{\ell=1}^6, \quad \chi(\ell) \in \{Q, K\}. \quad (57)$$



Here the six members are the four aligned/compact score and gradient layouts shared with the ordinary API plus the compact dQ/dK matrix operands. The layout maps  $\Pi_\ell$  change byte placement only; no member requantizes the fragment. The fixed representation also removes the adaptive seven-word head record from this specialization. Its forward scale hierarchy is

$$s_{\text{local}}^Q = s_{\text{local}}^K = 448, \quad g^Q = g^K = \frac{1}{448 \cdot 16}, \quad (58)$$

while the compact backward dQ/dK operands need no materialized scale tensors. All six payloads and the forward scale pages are bit-exact against the standalone fixed-scale publisher. This is an intentionally pure-only contract: adaptive FP4+FP8 continues to use the robust per-head record.

## 5.6 Projection-native MXFP4 V

For the BF16-rounded projection fragment, define the two-dimensional tile  $T_{u,v} = \{(t, d) : 32u \leq t < 32(u+1), 32v \leq d < 32(v+1)\}$ . The retained MXFP4 policy forms one E8M0 scale for the complete tile,

$$s_{u,v}^V = Q_{\text{E8M0}} \left( \frac{\max_{(t,d) \in T_{u,v}} |V_{R,t,d}|}{6} \right), \quad (59)$$

$$C_{t,d}^V = Q_{\text{E2M1}} \left( \frac{V_{R,t,d}}{s_{u,v}^V} \right), \quad \hat{V}_{t,d} = s_{u,v}^V C_{t,d}^V. \quad (60)$$

The scale is repeated into the thirty-two hardware  $1 \times 32$  scale records inside  $T_{u,v}$ . The same codes and scale are then written in both the feature-major layout required by forward PV and the row-major layout required by backward dP. Therefore

$$\mathcal{B}_{\text{bwd}}^V = \Pi_{SD}(\mathcal{B}_{\text{fwd}}^V), \quad (61)$$

with no independent row-scale selection after the transpose. This removes the BF16 V reread and standalone row-to-column packing from forward setup and makes the two MXFP4 consumers observe one represented V operator. The legacy  $1 \times 32$  policy remains available only as an explicit diagnostic.

The complete optional publication contract is

$$\mathcal{B}_{QKV} = \{C_Q, C_K, s^Q, g^Q, s^K, g^K, C^V, s^V, \mathcal{L}_{Q,\text{seq}}, \mathcal{L}_{K,\text{aligned}}, m_h^Q, m_h^K\} \cup \{Q_{\text{BF16}}, K_{\text{BF16}}, V_{\text{BF16}}\}_{\text{optional}}. \quad (62)$$

When every downstream consumer accepts  $\mathcal{B}_{QKV}$ , disabling the three BF16 stores avoids  $B_{SH}(192 + 192 + 128) \cdot 2 = B_{SH} \cdot 1024$  bytes of global publication.

## 5.7 Scale ownership at the composed D64 boundary

The projection epilogue also publishes E4M3 Q, K, and V for attention backward. These are deliberately independent of the forward FP4 multipliers  $m_h^Q$  and  $m_h^K$ . Let  $\lambda = 2^2 = 4$  be the fixed E4M3 publication factor and let  $G = dO$  denote the loss-scaled upstream gradient. Backward consumes

$$Q' = \lambda Q, \quad K' = \lambda K, \quad V' = \lambda V, \quad G' = \lambda G, \quad (63)$$



and compensates the score product with

$$\alpha' = \frac{\alpha}{\lambda^2} = \frac{\alpha}{16}. \quad (64)$$

The represented softmax is consequently unchanged. Applying the chain rule in encoded units gives

$$dV' = \lambda dV, \quad dP' = \lambda^2 dP, \quad dQ' = \lambda dQ, \quad dK' = \lambda dK. \quad (65)$$

If dynamic loss scaling contributes  $L$ , inverse RoPE and the QKV projection consume each attention-gradient field with  $1/(\lambda L)$ . Optional calibration gains  $c_Q, c_K, c_V$  remain field-local,

$$\bar{d}Q = \frac{c_Q}{\lambda L} R^\top dQ', \quad \bar{d}K = \frac{c_K}{\lambda L} R^\top dK', \quad \bar{d}V = \frac{c_V}{\lambda L} dV'. \quad (66)$$

The default is  $c_Q = c_K = c_V = 1$ . A forward probability approximation does not implicitly own these gains: any non-unit correction must be exported by the topology and validated per field. In particular, a historical common 0.632 multiplier is not inferred from the softmax mode because it suppresses Q/K/V gradient norms without changing the output-projection weight gradient.

## 5.8 Adaptive scale production remains a contract

Outside the fixed-scale pure specialization, the fused epilogue *consumes* the seven-word per-head scale record; it does not calculate Equation 13 from its distributed output tiles. A true same-launch amax/RMS reduction would need cross-CTA completion before earlier Q/K tiles could be finalized, defeating the simple streaming epilogue.

The current API consequently exposes three legal scale sources:

1. the standalone robust Q/K reduction, used for validation;
2. precomputed device-resident scales from an upstream producer or calibration policy;
3. a future learned-projection epilogue with a delayed, pipelined, or model-defined scale contract.

Current projection validation derives the record from BF16 reference Q/K before timing the projection hot stage. The measured epilogue is real, but this detail prevents interpreting its timing as a solved same-step adaptive scale reduction. An integrated model must state how  $m_h^Q$  and  $m_h^K$  are supplied.

## 5.9 Upstream low-precision dO contract

The backward also needs two MXFP4 orientations of the output gradient  $G = dO$  and the softmax centering term. For row blocks  $b_r$  and transposed column blocks  $b_c$ , the producer emits

$$C_{i,b_r}^{G,r} = Q_{E2M1} \left( G_{i,b_r} / s_{i,b_r}^{G,r} \right), \quad (67)$$

$$C_{d,b_c}^{G,c} = Q_{E2M1} \left( G_{d,b_c}^\top / s_{d,b_c}^{G,c} \right), \quad (68)$$

$$\delta_i = \sum_{d=1}^{128} O_{i,d} G_{i,d}. \quad (69)$$

One fused producer reads dO once and publishes both code/scale orientations and  $\delta$ . This is the upstream boundary used by the all-MX experiments; the separate row quantizer, column quantizer, and delta kernel remain only as a bit-exact diagnostic reference.

### 5.10 Probability replay boundary

Within one backward iteration, the retained route reuses the represented E4M3 probability produced for dV when forming dS rather than reevaluating the exponential. If  $C^P$  denotes that code,

$$\widehat{dS}_{ij} = \widehat{P}_{ij} \left( \widehat{dP}_{ij} - \delta_i \right), \quad \widehat{P}_{ij} = s_{ij}^P C_{ij}^P. \quad (70)$$

This is a local register/TMEM reuse contract, not an HBM cache. Caching a complete MXFP4 probability matrix across forward and backward would require

$$M_P = \frac{BHS^2}{2} + BH \left( \frac{S}{128} \right)^2 (32 \cdot 16) \text{ bytes} \quad (71)$$

for payload plus scale pages. At S8192/H8 this is approximately 272 MiB, so the write/read traffic overwhelms the measured local replay saving.

### 5.11 Reduction-width dispatch

Publication cost is controlled by projection reduction depth  $K$ , not by head count in isolation. With only four K256 steps at K1024, one consumer warpgroup cannot hide BF16 rounding, E2M1 conversion, transpose, and four global layouts behind the tensor producer. The separate 32-register packer spreads that work over a wider CTA grid. With twelve or more K256 steps, publication hides under the MMA pipeline and avoiding the BF16 HBM reread is decisive.

The legacy Q/K-only public dispatcher therefore selects

$$\text{publish}(K) = \begin{cases} \text{NVFP4 projection} + \text{parallel packer}, & K < 3072, \\ \text{fused register-epilogue publication}, & K \geq 3072. \end{cases} \quad (72)$$

Both paths return one `B300AdaptiveLowpOperands` object and identical layout bits.

The unified QKV entry point always uses the fused projection epilogue because V publication and optional BF16 suppression are part of the single producer contract. Its projection scratch aliases the Q/K code staging region with the V BF16-pair staging region: they are disjoint in lifetime. This reduces static scratch from 10,704 to 8,592 bytes without changing the hot backward kernel.

### 5.12 Projection backward

For the PyTorch weight convention in Equation 1, the input-gradient contribution is

$$dX = dQW_Q + dKW_K + dVW_V. \quad (73)$$

The first integration writes fully reduced BF16 gradients into one per-head buffer ordered as  $[dQ_{192}, dK_{192}, dV_{128}]$ . A persistent weight packing step stores  $[W_Q, W_K, W_V]$  in the matching order. One, two, or four head-contiguous GEMMs consume the buffer, selected by shape, without three standalone gradient tensors or an explicit concatenation.

The newer dQ path joins readiness to the projection reduction loop. Let  $r$  index a 256-row projection tile,  $h$  a head, and  $o$  a causal K/V-owner cluster. Each of the two CTAs in owner  $o$  publishes one

release increment only after all four reducer warps have retired their BF16 TMA reductions and ordered the async proxy:

$$a_{h,t} \leftarrow a_{h,t} + 1, \quad \text{ready}(h, t, r) \iff a_{h,t} \geq 2(r + 1), \quad (74)$$

$$t \in \{2r, 2r + 1\}. \quad (75)$$

The factor  $2(r + 1)$  is the exact number of contributing CTAs for either 128-row half of causal row pair  $r$ . A device-scope acquire poll is placed immediately before the first K64 slice of each head, not before the complete projection tile. The projection accumulator therefore evaluates

$$dX_{R,C} = \sum_{h=0}^{H-1} \sum_{j=0}^2 dQ_{R,h,64j:64(j+1)} W_{Q,C,h,64j:64(j+1)}^T, \quad (76)$$

waiting for head  $h$  only when its first  $j = 0$  slice reaches the load producer. This is a reduction-topology integration: readiness is part of the K loop, rather than a whole-row gate in front of GEMM.

Two FP32 TMEM accumulators alternate output tiles. While the epilogue drains slot  $p$  to BF16, the tensor warpgroup issues the next output into slot  $p \oplus 1$ . This removes the former MMA–epilogue serialization without allocating TMEM inside the attention kernel; the projection kernel owns all 512 of its own TMEM columns. The consumer remains at 80 registers with zero spills, and the attention specialization remains at 128 registers with zero spills.

A bounded early grid is necessary so projection does not displace the attention owner grid. For wide outputs, that bounded grid cannot expand when attention retires, so an experimental split computes a ready row prefix with the persistent consumer and a disjoint suffix with the vendor GEMM:

$$dX = \begin{bmatrix} \mathcal{P}_\theta(dQ_{0:R}, W_Q) \\ \text{GEMM}(dQ_{R:S}, W_Q) \end{bmatrix}. \quad (77)$$

The two streams write disjoint rows. At S8192/H8 the all-persistent route now slightly beats attention plus cuBLAS. H16/H24/H64 still favor the ordinary BF16 reduction followed by the vendor GEMM, so the public entry point shape-dispatches to that non-regressing fallback unless the pipelined route is explicitly forced for profiling.

### 5.13 Absorbing a hierarchical dQ sum

The next experiment removes the *completed* standalone dQ tensor from the projection contract. Partition K/V owners by parity and define two BF16 reduction lanes

$$dQ^{(\ell)} = \sum_{o: o \bmod 2 = \ell} dQ^{(o)}, \quad dQ = dQ^{(0)} + dQ^{(1)}. \quad (78)$$

The attention reducer publishes each owner into its compile-time-selected lane. Projection then exploits linearity:

$$dX_Q = (dQ^{(0)} + dQ^{(1)}) W_Q \quad (79)$$

$$= dQ^{(0)} W_Q + dQ^{(1)} W_Q. \quad (80)$$

Both lane products accumulate into the same FP32 TMEM output. No kernel ever constructs or consumes a global BF16 array equal to  $dQ$ . The ordinary one-lane producer is a separate template instantiation, so hierarchy indexing does not add a modulo or branch to the retained path.

The construction is numerically sound but does not improve time. At S8192/H8, attention plus cuBLAS measured 0.773824 ms and the two-lane consumer 0.805888 ms, a 32.064  $\mu$ s loss, with dX cosine 0.99999422. At S4096/H24, the corresponding times were 0.736000 ms and 1.063392 ms, a 327.392  $\mu$ s loss, with cosine 0.99999648. Both projection specializations compile at 80 registers, two barriers, and zero spills.

Equation 80 duplicates the complete weight MMA. A second implementation instead formed  $dQ^{(0)} + dQ^{(1)}$  in shared memory before one MMA. Scalar and packed-BF16 serial folds took approximately 2.68 and 2.33 ms. Moving the fold to idle producer warps and using explicit 128-bit shared transfers improved it to 0.860608 ms at S8192/H8, but remained slower than the two-MMA form. The tensor-native hierarchy is therefore kept only as an opt-in structural diagnostic.

This distinction matters: the experiment eliminates a completed BF16 dQ *tensor*, but not BF16 global partial traffic. The producer still emits the same number of values, split across two arrays, and projection pays either an additional weight product or an explicit fold. A winning hierarchy must also change the pre-publication representation:

$$\text{owner partials} \xrightarrow{\text{on-chip grouping or compact FP8/FP4}} \text{projection reduction.} \quad (81)$$

A winning path performs Equation 81 without a BF16 dQ round trip. An on-chip super-owner can reduce several K/V owners before one global write. Alternatively, compact low-precision partials can make repeated projection products proportional to the higher FP8/FP4 tensor rate. Reversing the whole kernel to q-centric ownership completes dQ locally but transfers the global reduction to the combined 320 dK/dV features rather than the 192 dQ features, so it is not the preferred next topology.

### 5.14 Stacked low-precision QKV-gradient handoff

The final experiment combines the three projection-gradient fields before one NVFP4 projection. Let  $R_t^\top$  apply inverse RoPE to one Q or K head at token  $t$ . For the two-lane diagnostic, the row presented to the quantizer is

$$G_t^{(2)} = \left[ R_t^\top (dQ_t^{(0)} + dQ_t^{(1)}) \mid R_t^\top dK_t \mid dV_t \right]. \quad (82)$$

The producer loads a K128 tile, assigns its two K64 halves to independent row groups, performs the dQ lane addition and inverse rotations in registers, and chooses one delayed-scale NVFP4 representation per K16 block:

$$s_{t,b}^G = Q_{E4M3} \left( \frac{\max_{k \in b} |G_{t,k}^{(2)}|}{6\bar{g}_G} \right), \quad (83)$$

$$C_{t,k}^G = Q_{E2M1} \left( \frac{G_{t,k}^{(2)}}{\bar{g}_G s_{t,b}^G} \right), \quad \hat{G}_{t,k} = \bar{g}_G s_{t,b}^G C_{t,k}^G. \quad (84)$$

With the correspondingly stacked persistent weight  $W_\Sigma = [W_Q^\top, W_K^\top, W_V^\top]^\top$ , one projection computes

$$\widehat{dX}_{t,j} = \bar{g}_G g_W \sum_b \text{MMA}_{E2M1} \left( C_{t,b}^G, C_{j,b}^W; s_{t,b}^G, s_{j,b}^W \right). \quad (85)$$

This form avoids a completed dQ array, but enabling per-tile readiness changes the attention schedule. Even a one-lane signaling control slows the attention kernel. The retained activation-only

variant therefore uses the ordinary unsigned BF16 dQ endpoint and replaces  $dQ^{(0)} + dQ^{(1)}$  in Equation 82 by completed  $dQ$ . It still fuses inverse RoPE, three-field layout conversion, local scaling, and packing before the single NVFP4 projection. It does *not* remove BF16 dQ materialization, and it does not provide the BF16 QKV tensor required by the projection weight-gradient GEMM. The two contracts are therefore reported separately rather than treating the activation-only result as a complete training step.

### 5.15 Compact NVFP4 dQ handoff

A second implementation changes the representation at the final attention–projection boundary without changing the owner reduction. Let  $D = dQ_{\text{complete}}$  be the private BF16 reduction scratch and let  $\bar{g}_D$  be a device-resident NVFP4 decode scale supplied by calibration or a previous step. For each K16 block  $b$ , the producer forms

$$s_{i,b}^D = Q_{\text{E4M3}} \left( \frac{\max_{k \in b} |D_{ik}|}{6\bar{g}_D} \right), \quad (86)$$

$$C_{ik}^D = Q_{\text{E2M1}} \left( \frac{D_{ik}}{\bar{g}_D s_{i,b}^D} \right), \quad \hat{D}_{ik} = \bar{g}_D s_{i,b}^D C_{ik}^D. \quad (87)$$

The matrix-wide amax pass is absent from this hot path. With a persistently packed projection weight  $\widehat{W}_{jk} = g_W s_{j,b}^W C_{jk}^W$ , the consumer evaluates

$$\widehat{dX}_{Q,ij} = \bar{g}_D g_W \sum_b \text{MMA}_{\text{E2M1}} (C_{i,b}^D, C_{j,b}^W; s_{i,b}^D, s_{j,b}^W). \quad (88)$$

The public result contains  $\widehat{dX}_Q$ , dK, and dV; standalone dQ is not returned. The generic NVFP4 consumer uses M256×N256 output tiles, K256 reduction steps, a two-CTA cluster, FP32 TMEM accumulation, and one BF16 output rounding. It compiles at 86 registers with two barriers and no spills.

This is a compact *final-tile* handoff, not yet a compact owner-partial reduction. The unclustered attention owners still require an additive BF16 TMA target, so  $D$  exists as private scratch until the last owner completes:

$$\{dQ^{(o)}\}_o \xrightarrow{\text{BF16 TMA reduction}} D \xrightarrow{Q_{\text{NV4}}} (C^D, s^D, \bar{g}_D) \xrightarrow{\text{NV4 projection}} \widehat{dX}_Q. \quad (89)$$

It nevertheless measures whether compact projection consumption can repay its conversion. Delayed scale overestimates from  $1\times$  through  $8\times$  the same-step ideal preserve approximately 0.99098 cosine because the local K16 scales renormalize the blocks. Underestimates clip:  $0.5\times$ ,  $0.25\times$ , and  $0.125\times$  give approximately 0.9905, 0.9849, and 0.9637 cosine. Consequently a conservative delayed policy is viable; a winning fused owner scheme must still publish tile-ready codes or quantize inside a super-owner.

### 5.16 The new ceiling

Unified projection publication removes three previously exposed costs:

$$\text{BF16 Q/K/V HBM publication} \rightarrow \text{Q/K/V HBM reread} \rightarrow \text{standalone pack/transpose/quantize}. \quad (90)$$

It does not remove the global dQ reduction. A timing-only attention run that suppressed dQ publication reached approximately 0.693 ms at S8192/H8, versus 0.746 ms for the direct-BF16 endpoint. The head-pipelined route now participates in readiness and reduction scheduling, but still reads the globally materialized BF16 sum. Reaching the remaining floor requires projection backward to absorb the hierarchical sum before that publication, not merely observe it earlier. The two-lane experiment shows more precisely that absorbing only the *last addition* is also insufficient: owner grouping or compact publication must remove the BF16 producer traffic at the same time.

## 6 Performance and Accuracy

### 6.1 Measurement boundary

All headline backward rows are causal B1 with  $D_{QK} = 192$  and  $D_V = 128$  on GPU 0. The copied BF16 V382 mode and low-precision mode run through the same extension and scheduler family. Table 5 uses five warmups and 21 rotated CUDA-event samples with seed 2026081201. Operand preparation is outside the standalone backward timing; projection tables state explicitly whether BF16-to-NVFP4 activation preparation is charged.

These are implementation medians, not peak-throughput estimates. The result does not include an optimizer step, projection weight-gradient GEMM, or full training iteration.

### 6.2 Standalone backward

Reducing dQ in BF16 and returning it directly gives the strongest deployable backward endpoint:

**Table 5:** Retained adaptive backward versus the copied BF16 control. Lower is better.

| S     | H  | BF16 control (ms) | adaptive BF16 dQ (ms) | improvement |
|-------|----|-------------------|-----------------------|-------------|
| 4096  | 24 | 0.704480          | 0.624832              | 11.31%      |
| 8192  | 8  | 0.868864          | 0.746432              | 14.09%      |
| 8192  | 24 | 2.207200          | 1.888768              | 14.43%      |
| 8192  | 64 | 5.560512          | 4.761888              | 14.36%      |
| 16384 | 24 | 7.743616          | 6.471232              | 16.43%      |
| 16384 | 64 | 20.273727         | 17.068480             | 15.81%      |

The FP32-return API uses the same BF16 reduction and adds one widening pass. Skipping that pass saves 2.0–5.0% relative to the FP32-return adaptive route. Changing the reduction payload itself from FP32 to BF16 accounts for another 0.5–5.2%, with the benefit increasing at long sequence.

### 6.3 Calibrated and outlier accuracy

Table 6 compares direct BF16 outputs to the copied BF16 control. Values are component cosine, not an aggregate dominated by dV.

The robust scale rule in Equation 14 matters outside the calibration distribution. On a one-percent sparse Q/K outlier test:

**Table 6:** Calibrated component cosine for the direct-BF16 endpoint.

|  | S     | H  | dQ       | dK       | dV       |
|--|-------|----|----------|----------|----------|
|  | 4096  | 24 | 0.991447 | 0.991379 | 0.999356 |
|  | 8192  | 8  | 0.991533 | 0.991557 | 0.999378 |
|  | 8192  | 24 | 0.991435 | 0.991484 | 0.999360 |
|  | 8192  | 64 | 0.991478 | 0.991487 | 0.999354 |
|  | 16384 | 24 | 0.991560 | 0.991526 | 0.999355 |
|  | 16384 | 64 | 0.991447 | 0.991452 | 0.999345 |

**Table 7:** Sparse-outlier accuracy of the retained adaptive route.

|  | S    | H  | dQ cosine | dK cosine | dV cosine |
|--|------|----|-----------|-----------|-----------|
|  | 4096 | 24 | 0.962695  | 0.962139  | 0.999070  |
|  | 8192 | 8  | 0.961676  | 0.963121  | 0.998915  |

The earlier global-amax adaptive rule reached only about 0.753/0.757 dQ/dK cosine at S4096/H24. The robust per-head rule improves the outlier regime without changing the three score instructions, TMEM map, or calibrated result. It still clips information and is not a proof of training convergence.

## 6.4 Learned projection publication

Table 8 charges the current standalone conversion of BF16 X into the NVFP4 activation operand. NVFP4 weights are treated as persistent and prepared once. The low-precision column uses the automatic publication policy in Equation 72.

**Table 8:** Q/K learned-projection plus backward-layout production. The BF16-input column includes current activation preparation.

| Shape           | BF16 GEMM + pack (ms) | BF16 input to NVFP4 auto (ms) | speedup |
|-----------------|-----------------------|-------------------------------|---------|
| S8192/H8/K1024  | 0.163712              | 0.142768                      | 1.147×  |
| S4096/H24/K3072 | 0.326704              | 0.206096                      | 1.585×  |
| S4096/H64/K8192 | 1.553184              | 0.515616                      | 3.012×  |

If X already arrives in NVFP4, the corresponding hot-stage speedups are 1.473×, 1.933×, and 3.342×. Q/K cosine against the BF16 learned projection is approximately 0.99097–0.99098. Fused versus separate E2M1 publication is bit-exact.

The dispatcher removes the shallow-K epilogue regression:

The dominant gain is real NVFP4 tensor work and reduced A/B traffic. At deep K, the fused epilogue adds a movement win by avoiding the BF16 Q/K reread and standalone transpose. At K1024, the wider parallel packer is faster than placing all publication on the persistent GEMM’s one consumer warpgroup. These two tables describe the legacy Q/K-only API.

The unified QKV producer adds projection-native MXFP4 V and allows all three BF16 stores to be suppressed. Table 10 compares it with three BF16 GEMMs followed by the standalone Q/K packer.



**Table 9:** Automatic projection publication versus forcing the other route.

| Shape           | auto route | auto (ms) | forced other (ms) | advantage |
|-----------------|------------|-----------|-------------------|-----------|
| S8192/H8/K1024  | separate   | 0.111024  | 0.122368          | 1.102×    |
| S4096/H24/K3072 | fused      | 0.169984  | 0.181712          | 1.069×    |
| S4096/H64/K8192 | fused      | 0.462560  | 0.626592          | 1.355×    |

Input and weight NVFP4 operands are already resident in all columns so that the table isolates projection and publication policy.

**Table 10:** Unified QKV projection and native forward/backward publication.

| Shape           | BF16 + QK pack<br>(ms) | unified + BF16<br>(ms) | speedup | unified no BF16<br>(ms) | speedup |
|-----------------|------------------------|------------------------|---------|-------------------------|---------|
| S2048/H8/K1024  | 0.149264               | 0.088688               | 1.683×  | 0.077440                | 1.927×  |
| S4096/H24/K4096 | 0.463744               | 0.328816               | 1.410×  | 0.289552                | 1.602×  |

Projection Q/K/V cosine is 0.99096–0.99098. Consuming the native operands in causal forward gives output cosine 0.990722 at S2048/H8 and 0.991317 at S4096/H24. All four backward Q/K layouts are bit-exact against their standalone publishers. Under the retained 2-D policy, the forward- and backward-oriented V payloads are exact transposes and their repeated scale pages select the same  $32 \times 32$  tile scale. The legacy standalone V publishers select independent  $1 \times 32$  scales, so they are intentionally no longer a byte-level reference. The forward Q/K tensors and backward compact tensors alias the same allocation.

The storage reduction is material at long sequence. Sharing the compact Q/K payload avoids  $BHS \cdot 192$  bytes: 12 MiB at S8192/H8 and 18 MiB at S4096/H24. Suppressing BF16 Q/K/V publication avoids a further 64 MiB and 96 MiB respectively. The projection producer also reduces its static shared scratch by 2,112 bytes, from 10,704 to 8,592 bytes. This scratch saving does not change the already near-full shared-memory allocation of the backward kernel.

## 6.5 Two-dimensional scale-contract audit

The composed D64 path was re-audited with the representation rules in Equations 38 and 60. Packing a synthetic weight and independently packing its transpose produces exactly the transposed E2M1 codes, matching E4M3 local scales, and the same global scale under the  $16 \times 16$  policy. The  $1 \times 16$  policy does not provide that invariant.

Sharing scales over two dimensions is stricter and therefore slightly less accurate for a single isolated tensor. On the fixed D64 projection audit, learned-projection cosine changes from 0.990916 with  $1 \times 16$  weights to 0.989259 with  $16 \times 16$  weights; relative L2 changes from 0.13483 to 0.14630. Decoded V cosine changes from 0.993560 with independently selected  $1 \times 32$  row/column scales to 0.993031 with one  $32 \times 32$  scale; relative L2 changes from 0.11332 to 0.11789. The retained reason for the two-dimensional policy is not a better one-shot cosine: it is transpose consistency when fprop and bprop actually consume the two orientations of the same low-precision representation. That benefit is only partially active in the selected hybrid. Output projection does consume both orientations, but QKV projection dgrad remains BF16 and attention backward consumes FP8 V.



Consequently, applying one 2-D switch to every learned weight and to  $V$  also charges coarser forward quantization where no transposed FP4 consumer reuses it. The policy remains useful for the all-FP4 ladder; the hybrid requires separate QKV/output-weight switches rather than a blanket 2-D default.

The E8M0 exponent selector must likewise follow the scale geometry. On the fixed causal- $V$  activation audit, the  $1 \times 32$  policy minimizes attention relative L2 at 0.11409 when it rounds up at normalized  $\text{amax}$  1.20; ordinary nearest-power rounding (the 1.50 midpoint) gives 0.12818. For a shared  $32 \times 32$  tile the ordering reverses: nearest-power rounding gives 0.12439, versus 0.12745 with the 1.20 rule. The retained publisher therefore uses a BF16-exact 1.203125 cutoff for  $1 \times 32$  and ties-to-even nearest-power rounding for  $32 \times 32$ . A staged binary matched the independent payload and scale emulator bit-for-bit for both geometries, with and without causal interleaving, and passed the unified projection-layout validator. These are one-activation representation checks, not convergence measurements. In the matched sixteen-layer diagnostic, the new selector improves first-layer attention-branch relative L2 from 0.37686 to 0.37438, but the final sampled logit changes are mixed and negligible: cosine changes from 0.68707 to 0.68597 while relative L2 changes from 0.79662 to 0.79494. Selector rounding therefore does not explain the much larger optimization-proxy regression.

The implementation cost is below whole-step measurement noise. In matched 16-layer S4096/H32 GQA runs, the low-precision step measured 76.929 ms with legacy one-dimensional scales and 76.931 ms with two-dimensional scales. The two-dimensional run used independent Q/K FP4 multipliers 2.25 and 2.0 and unit field gains. Against an 83.304-ms BF16 step in that run, it measured  $1.083\times$  end-to-end speedup. This short synthetic-token comparison is a systems measurement; its 16-layer sampled-logit cosine of 0.6989 is not a convergence result.

The backward scale audit also found and removed a stale common gain. The previous MXFP4-PV comparison silently inferred 0.632 from its forward softmax mode, yielding initial Q/K weight-gradient norm ratios near 0.593 and a  $V$  ratio near 0.425. With Equation 66 and unit gains, the corresponding Q/K/ $V$  ratios are approximately 0.954/0.937/0.961. Directional error remains after sixteen approximate layers, however, and norm parity does not imply a more accurate gradient. If  $r = \|\hat{g}\|/\|g\|$  and  $c$  is the cosine to the BF16 gradient, applying gain  $a$  gives

$$\frac{\|a\hat{g} - g\|_2}{\|g\|_2} = \sqrt{1 + (ar)^2 - 2arc}. \quad (91)$$

For  $c \approx 0.3$ , forcing  $ar \approx 1$  raises relative error to about 1.18. Thus the unit-gain endpoint removed a norm confound but amplified a poorly aligned direction. It is an audit point, not a calibrated training policy. The optional NVFP4 QKV-dgrad consumer now folds the same three decode factors into its E4M3 metadata. It restores the expected norms but is not selected: its measured backward is 46.11 ms versus 43.89 ms for the BF16 projection dgrad route.

No random Hadamard transform or stochastic rounding is charged in the selected route. Forward activations use  $1 \times 16$  RNE, learned weights use  $16 \times 16$  RNE, and QKV projection dgrad plus both weight-gradient GEMMs remain BF16. RHT/SR becomes relevant only if those gradient operands are actually retained in FP4; adding it around the slower diagnostic NVFP4 dgrad would be fake quantization overhead rather than a numerical fix for the winner.

A controlled 24-round comparison then put both PV routes on this corrected contract. Table 11 reports medians for the synthetic 16-layer S4096 Llama-1.2B training proxy.

The two low-precision backward times are indistinguishable at this scale; the MXFP4 route's additional 2.43-ms whole-step gain comes from forward PV. The corresponding sampled-logit cosines

**Table 11:** End-to-end D64 route comparison with 2-D learned weights, 2-D MXFP4 V, independent Q/K multipliers, and unit Q/K/V backward gains.

| Route                   | forward<br>(ms) | backward<br>(ms) | optimizer<br>(ms) | step<br>(ms) | speedup<br>vs BF16 |
|-------------------------|-----------------|------------------|-------------------|--------------|--------------------|
| BF16 CuTE               | 27.718          | 45.596           | 10.726            | 84.050       | 1.000×             |
| NVFP4 QK + MXFP4 PV     | 22.466          | 44.239           | 10.732            | 77.428       | 1.086×             |
| NVFP4 QK + exact FP8 PV | 24.884          | 44.233           | 10.733            | 79.860       | 1.052×             |

are 0.69348 for MXFP4 PV and 0.77821 for exact FP8 PV. Q/K/V first-layer weight-gradient cosine is 0.27691/0.32003/0.39347 for MXFP4 and 0.44378/0.45383/0.68284 for FP8, while Q/K gradient norm ratios remain near 0.94–0.96 in both routes. Therefore the corrected difference is primarily directional PV error, not the removed common-gain bug. The routes share the same QK policy only at the first layer: their distinct PV outputs perturb the residual stream, so subsequent layers quantize different hidden states.

All routes remained finite in the short 24-step optimization proxy. The last-cycle median losses were 4.281 for BF16, 5.797 for MXFP4 PV, and 5.344 for exact FP8 PV. A later audit found that the compile warmup performed one unreported optimizer update, so these values must not be used to select a training policy. The speed measurements remain systems measurements, but the apparent loss ordering motivated the controlled ablation below.

## 6.6 Post-hoc loss-regression ablation

The endpoint above changed four policies together: learned weights from 1-D to 2-D scales, V from 1-D to 2-D scales, Q/K multipliers from 1.75/1.75 to 2.25/2.0, and the inferred backward gain from 0.632 to 1.0. An initial eight-run factorial held seeds, token batches, route order, and optimizer settings fixed. Before discovery of the hidden warmup update, that factorial attributed 0.596 of the 0.691 mean MX–BF16 loss-gap regression (86.3%) and 0.941 of the 1.037 last-four regression (90.8%) to the gain change. Those endpoint losses are not themselves valid policy evidence, but the factor ranking motivated the corrected controls below. Changing Q/K multipliers alone is neutral within run noise. Two-dimensional learned weights reduce initial sampled-logit cosine by about 0.029 in both PV routes and modestly worsen the loss proxy; two-dimensional V has a much smaller deterministic effect. Gain changes initial logits by exactly zero.

The mechanism follows Equation 91. In the sixteen-layer endpoint, Q/K gradient cosine is only about 0.28–0.32 for MX and 0.44–0.45 for FP8. Raising MX Q/K/V norm ratios from roughly 0.59/0.60/0.44 to 0.94/0.95/0.75 therefore increases relative L2 even though the norms look healthier. By contrast, the isolated one-layer audit gives Q/K/V weight-gradient cosine 0.89/0.89/0.91 for MX and 0.93/0.92/0.93 for FP8. The local stitch operation is bit-exact for Q, K, and V, while the attention kernel itself gives  $dQ/dK$  cosine near 0.997 and  $dV$  near 0.9995. The poor sixteen-layer direction is therefore accumulated forward-trajectory error, not a local backward layout failure.

The warmup was changed to allocate and compile fused AdamW state at zero learning rate, reset its moments and step counters, and leave parameters unchanged. Under this corrected harness, audited initial loss equals recorded round-zero loss exactly. Table 12 shows that the four-batch memorization proxy improves monotonically as the common gain tends toward zero.

The corrected matched endpoints independently reproduce the original factor ranking. Moving from

**Table 12:** Corrected 24-step repeated-batch proxy. Entries are mean relative loss differences to each run’s BF16 route; negative is lower on this proxy.

| Common gain | MX mean | MX last four | FP8 mean | FP8 last four |
|-------------|---------|--------------|----------|---------------|
| 0.02        | -25.29% | -45.93%      | -37.29%  | -74.12%       |
| 0.05        | -23.05% | -42.56%      | -34.27%  | -67.30%       |
| 0.10        | -19.40% | -35.77%      | -28.90%  | -58.42%       |
| 0.20        | -14.50% | -25.35%      | -23.36%  | -46.68%       |
| 0.40        | -7.36%  | -11.23%      | -15.22%  | -33.56%       |
| 0.50        | -4.35%  | -2.06%       | -11.85%  | -26.11%       |
| 0.632       | 2.05%   | 10.91%       | -5.16%   | -9.49%        |
| 1.00        | 14.27%  | 39.15%       | 9.58%    | 26.37%        |

0.632 to 1.0 worsens the mean and last-four relative gaps by 12.22 and 28.25 percentage points for MXFP4 PV, and by 14.74 and 35.86 points for FP8 PV. The BF16 last-cycle median differs by only 0.0078 between those separately scheduled runs.

This monotone result is a harness pathology, not a low-precision win. A gain of 0.02 nearly freezes Q/K/V projection-weight gradients and the attention contribution to upstream  $dX$ , while residual, MLP, embedding, and output projection paths continue to memorize four repeated random sequences. The proxy therefore rewards routing learning around attention and cannot select a production gain.

A unique-batch control confirms that interpretation. When the same 24 steps consume 24 independently generated random sequences, every route begins and ends at BF16-rounded loss 12.1875 and mean loss remains approximately 12.18. Changing the gain from 0.02 to 1.0 then changes the mean relative route gap by only 0.0001 percentage points for MXFP4 PV and 0.0427 points for FP8 PV. On the four-batch cycling proxy, the same endpoint comparison changes the mean gap by 39.56 and 46.87 points. The dramatic gain curve is therefore a repeated-example memorization effect, not evidence of model convergence or generalization.

A field ablation makes the failure more specific. Starting from all fields at 0.02, raising Q or K to 0.5 slightly improves both routes, and raising both improves the mean absolute loss gap by 0.163 (MX) and 0.188 (FP8). Raising V alone worsens it by 1.513 and 1.850 respectively. A complete  $2 \times 2$  V-gain factorial separates V’s projection-weight update from its contribution to upstream  $dX$ . Relative to the low/low corner, raising only V-to- $dX$  worsens the mean absolute loss gap by 1.513 (MX) and 1.888 (FP8), whereas raising only the V-weight gain changes it by just 0.051 and  $-0.020$ . The high/high corner reproduces the V-field regression. This is a route-common property of this proxy—the FP8 effect is larger—rather than evidence for an MXFP4-specific V-layout defect. The post-attention stitch was independently checked bit-exactly, and isolated  $dV$  cosine exceeds 0.9993, which rules out a gross transposition or decode-scale bug. The factorial establishes that V-to- $dX$  controls this proxy response; it does not establish that this path uniquely controls real-training convergence.

The actionable conclusion is twofold. First, no gain from this synthetic curve should be promoted; gain and learning rate must be co-tuned on real, non-repeated tokens with held-out loss. Second, numerical work should target forward trajectory fidelity—especially the probability/V path—rather than normalizing already misaligned full-model gradients.

## 6.7 RoPE-aware projection and composed decoder boundary

Table 13 isolates full-head pair-native RoPE at S4096/H24/K3072. “Separate” runs the same projection and then a strong in-place BF16 Q/K rotary kernel. The retained fused implementation uses packed BF16 arithmetic in the projection epilogue.

**Table 13:** Median component timing for pair-native RoPE publication.

| Operation                              | median (ms) |
|----------------------------------------|-------------|
| unrotated unified projection           | 0.341280    |
| projection with fused packed-BF16 RoPE | 0.432384    |
| projection then standalone RoPE        | 0.448512    |
| standalone forward RoPE only           | 0.114304    |
| standalone inverse-RoPE gradient only  | 0.114272    |

Fusion saves 16.128  $\mu$ s against the separate forward transform, but it does not make RoPE free: the rotation extends the projection consumer path by 91.104  $\mu$ s. The packed-BF16 result has Q/K cosine 0.9999978 and relative L2 approximately 0.00210 against an FP32-FMA epilogue reference; its maximum absolute difference is 0.00390625. All four E2M1 layouts are byte-exact against standalone publication of the same rotated BF16 tensors, and V is bit-exact. End-to-end Q/K projection cosine against dense standard-order RoPE is approximately 0.990966, so the incremental rotary rounding is small relative to the native NVFP4 projection error.

The inverse-gradient handoff has a different ownership problem. The generic NVFP4 quantizer assigns one complete 128-column row to each thread, whereas the standalone inverse transform assigns adjacent rotary pairs across a warp. Table 14 measures the retained direct-table specialization at the same S4096/H24 shape. Payloads, E4M3 scale pages, and the optional BF16 republication are bit-exact with the separate reference.

**Table 14:** Median inverse-RoPE and delayed-scale NVFP4 handoff timing at S4096/H24.

| Operation                                | median (ms) |
|------------------------------------------|-------------|
| standalone inverse RoPE                  | 0.114592    |
| NVFP4 pack with precomputed global scale | 0.059968    |
| separate inverse RoPE then NVFP4 pack    | 0.159712    |
| fused handoff, no BF16 republication     | 0.156992    |
| fused handoff with BF16 republication    | 0.156320    |

The no-publication path saves 2.720  $\mu$ s, or 1.70%, and compiles at 122 registers with no spills. The publication form compiles at 153 registers without spills and saves 3.392  $\mu$ s, or 2.12%; its asynchronous BF16 store is overlapped by the packing tail. Cooperative table staging was also tested: despite coalescing the cos/sin reads, its longer live range raised the kernels to 182–250 registers and made them substantially slower. Inverse RoPE is therefore not the missing source of a forward-sized end-to-end gain; eliminating the BF16 gradient boundary itself remains necessary.

A rotated 31-sample comparison in the composed harness confirms the system effect rather than extrapolating it from component times:

Relative to the same low-precision path with a separate inverse transform, fusion reduces activation-

**Table 15:** Fused inverse-RoPE handoff at S4096/H24/K4096 after seven warmups.

| Scope                | BF16 (ms) | separate (ms) | fused (ms) | fused vs BF16 |
|----------------------|-----------|---------------|------------|---------------|
| activation gradients | 2.188992  | 1.701568      | 1.693664   | 1.292×        |
| full training        | 2.620640  | 2.138720      | 2.125952   | 1.233×        |

scope time by  $7.904 \mu\text{s}$  (0.46%) and full training time by  $12.768 \mu\text{s}$  (0.60%). The fused BF16 republication makes the projection weight-gradient result bit-identical to the separate path, so the training row does not omit a required consumer.

Table 16 measures the dense causal decoder-attention boundary with QKV projection, fused RoPE, retained FP4 forward, output projection, output-projection dgrad, attention backward, inverse RoPE, and QKV projection dgrad. “Activation” omits projection weight gradients; “training” adds both projection weight-gradient GEMMs. Persistent weight operands, rotary tables, pair-native weight order, adaptive head records, and delayed global scales are cached outside timing. The selected column is the upstream-native delayed-scale NVFP4 route before the inverse-handoff fusion isolated in Table 15.

**Table 16:** RoPE-aware composed boundary on GPU 0. Times are medians over 13 samples after four warmups.

| S     | H  | K    | scope      | BF16 (ms) | low precision (ms) | speedup |
|-------|----|------|------------|-----------|--------------------|---------|
| 4096  | 24 | 4096 | activation | 2.195808  | 1.701600           | 1.290×  |
| 4096  | 24 | 4096 | training   | 2.627328  | 2.144448           | 1.225×  |
| 8192  | 24 | 4096 | activation | 5.037664  | 4.206208           | 1.198×  |
| 8192  | 24 | 4096 | training   | 5.875584  | 5.039552           | 1.166×  |
| 4096  | 64 | 8192 | activation | 7.119584  | 4.401792           | 1.617×  |
| 4096  | 64 | 8192 | training   | 9.208512  | 6.464608           | 1.424×  |
| 8192  | 64 | 8192 | activation | 16.682880 | 11.233536          | 1.485×  |
| 8192  | 64 | 8192 | training   | 20.868896 | 15.431264          | 1.352×  |
| 16384 | 64 | 8192 | activation | 44.307362 | 33.336990          | 1.329×  |
| 16384 | 64 | 8192 | training   | 52.857922 | 42.032127          | 1.258×  |

Across these rows, output-projection  $Y$  cosine is 0.97241–0.97397,  $dX$  is 0.98115–0.98146,  $dQ$  is 0.95368–0.96084,  $dK$  is 0.95348–0.95956, and  $dV$  is 0.99006–0.99045. The boundary is intentionally not advertised as a stock Llama drop-in: it retains  $D_{QK} = 192$ , equal Q/K/V head counts, and lacks grouped-query attention and native  $D_{QK} = 128$  kernels. It is nevertheless a RoPE-correct dense decoder attention measurement rather than the earlier RoPE-free proxy. The separate inverse transform costs about 0.115 ms at the component shape. The fused handoff recovers only 2.720–3.392  $\mu\text{s}$  within the isolated producer because its row-owned quantizer mapping makes rotary-table access strided. The composed gain is 0.46–0.60% at S4096/H24/K4096. A larger gain requires the projection consumer to accept the completed reduction tile directly, not merely to fuse arithmetic into a later global-memory quantizer.

## 6.8 Stacked QKV-gradient handoff

Table 17 compares the two variants in Equations 82–85 against the previously selected interleaved-gradient route. All three rows use the retained FP4 forward and FP4+FP8 backward. Times are complete activation-gradient boundaries, not isolated packers.

**Table 17:** Stacked QKV-gradient publication at S4096/H24/K3072. Medians use 21 rotated samples after eight warmups.

| Route                                              | time (ms) | result versus selected |
|----------------------------------------------------|-----------|------------------------|
| interleaved QKV gradient + fused inverse-RoPE pack | 1.578336  | baseline               |
| unsigned one-lane stacked QKV pack + projection    | 1.566048  | 0.78% faster           |
| signed two-lane hierarchical pack + projection     | 1.635936  | 3.65% slower           |

The one-lane output has dX cosine 0.981765 against the complete BF16 chain; against the selected low-precision dX its cosine is 0.99999994 and relative L2 is  $6.45 \times 10^{-5}$ . dK and dV are bit-identical to the selected attention outputs before the common projection quantization. The two-lane form is numerically valid but loses because tile-ready signaling adds approximately  $37 \mu s$  even in a one-lane control and roughly  $48 \mu s$  for the actual hierarchy. Its arithmetic savings do not repay that perturbation. The unsigned form is therefore retained only for activation dgrad. Full training retains the interleaved BF16 publication because the QKV weight-gradient GEMM requires those values.

## 6.9 Useful throughput and MFU

For one causal B1 layer, let  $K_m$  be model width and  $D_\Sigma = 2D_{QK} + D_V = 512$ . The harness counts useful dense and causal matrix FLOPs as

$$F_{\text{alg}} = 4SK_m H(D_\Sigma + D_V) + 3HS^2(D_{QK} + D_V) + \mathbf{1}_{\text{wgrad}} 2SK_m H(D_\Sigma + D_V). \quad (92)$$

Elementwise softmax, RoPE, scaling, packing, and reductions are excluded from this numerator. Both routes therefore have the same useful-work count, so the MFU multiplier is exactly the measured speedup. For an absolute common denominator, Table 18 uses 2.5 PFLOP/s dense BF16-equivalent throughput per GPU, derived from NVIDIA’s published two-GPU GB200 superchip specification and its dense-is-half-sparse note [5]. This is an algorithmic MFU, not a claim that every operation executes in BF16 or that a mixed FP4/FP8 kernel should be divided by the FP4 peak.

**Table 18:** Complete FP4-forward plus FP4+FP8-backward boundary on GPU 0. Medians use 11 rotated samples after five warmups. Training includes QKV and output projection weight gradients.

| S    | H  | $K_m$ | scope      | BF16 / lowp (ms)  | speedup        | MFU BF16 $\rightarrow$ lowp |
|------|----|-------|------------|-------------------|----------------|-----------------------------|
| 4096 | 24 | 3072  | activation | 1.9845 / 1.5696   | 1.264 $\times$ | 23.37% $\rightarrow$ 29.55% |
| 4096 | 24 | 3072  | training   | 2.2975 / 1.9114   | 1.202 $\times$ | 26.92% $\rightarrow$ 32.36% |
| 8192 | 16 | 2048  | activation | 3.0048 / 2.6881   | 1.118 $\times$ | 22.87% $\rightarrow$ 25.56% |
| 8192 | 16 | 2048  | training   | 3.3056 / 3.0152   | 1.096 $\times$ | 24.95% $\rightarrow$ 27.35% |
| 8192 | 24 | 4096  | activation | 5.0545 / 4.0739   | 1.241 $\times$ | 28.55% $\rightarrow$ 35.42% |
| 8192 | 24 | 4096  | training   | 5.8852 / 4.9502   | 1.189 $\times$ | 31.53% $\rightarrow$ 37.48% |
| 4096 | 64 | 8192  | activation | 7.1005 / 4.1286   | 1.720 $\times$ | 36.78% $\rightarrow$ 63.25% |
| 4096 | 64 | 8192  | training   | 9.1851 / 6.2505   | 1.469 $\times$ | 40.40% $\rightarrow$ 59.37% |
| 8192 | 64 | 8192  | activation | 16.7172 / 10.8004 | 1.548 $\times$ | 36.17% $\rightarrow$ 55.99% |
| 8192 | 64 | 8192  | training   | 20.8910 / 15.0497 | 1.388 $\times$ | 39.47% $\rightarrow$ 54.79% |

The activation rows use the unsigned stacked handoff; the training rows use the weight-gradient-capable interleaved handoff. Both use the FP4 FA4 forward. A BF16-forward hybrid is faster on several shapes, but is not used in this table because the measured question is the complete



FP4-forward system. Across the same five shapes, training-scope QKV weight-gradient cosine is 0.99007–0.99049, dX is 0.98092–0.98181, dQ/dK are 0.95573–0.95926/0.95523–0.95708, and dV is 0.99008–0.99050.

### 6.10 Multi-step convergence proxy

Operator cosine does not establish optimization behavior. A separate teacher/student experiment therefore updates Q, K, V, and output-projection weights for 500 streamed synthetic activation batches. Both students begin from the same 20%-perturbed teacher weights and use AdamW with learning rate  $2 \times 10^{-5}$ , loss scaling  $2^{20}$ , and identical data order. The low-precision student refreshes packed weight operands after every update and uses the complete FP4 forward and FP4+FP8 backward. Validation uses a held-out activation matrix.

The FP4 forward creates a nonzero teacher quantization floor. Let  $L_{\text{floor}}$  be the native low-precision validation loss obtained by the exact teacher weights and define reducible excess

$$E_{\text{lowp}} = L_{\text{lowp}} - L_{\text{floor}}. \quad (93)$$

Table 19 reports both raw loss and progress above that measured floor. “Improvement retained” is the low-precision absolute loss decrease divided by the BF16 decrease. The final column is  $(E_T/E_0)/(L_T^{16}/L_0^{16})$ ; values below one mean the reducible low-precision error decayed by a larger fraction.

**Table 19:** Five-hundred-step S4096/H24/K3072 teacher/student convergence proxy.

| Seed       | BF16 $L_0 \rightarrow L_T$    | lowp $L_0 \rightarrow L_T$    | floor   | improvement retained | normalized remaining ratio |
|------------|-------------------------------|-------------------------------|---------|----------------------|----------------------------|
| 2026081417 | 0.08143 $\rightarrow$ 0.05766 | 0.14579 $\rightarrow$ 0.12212 | 0.09760 | 0.996                | 0.718                      |
| 2026081431 | 0.08319 $\rightarrow$ 0.05750 | 0.14465 $\rightarrow$ 0.12141 | 0.09506 | 0.905                | 0.769                      |
| 2026081447 | 0.08139 $\rightarrow$ 0.05675 | 0.14071 $\rightarrow$ 0.11861 | 0.09346 | 0.897                | 0.763                      |

The low-precision route retains 93.2% of BF16’s absolute validation-loss improvement on average (89.7–99.6%). Its normalized excess remaining is 0.524 versus 0.699 for BF16’s normalized raw loss. This is evidence that the mixed backward does not prevent optimization in this block-level test. It is not evidence of equal final loss: native low-precision loss remains higher because of the 0.093–0.098 forward floor, and evaluating the learned low-precision weights through the common BF16 forward leaves final loss 25.5% higher on average. The experiment is not language-model pretraining, does not include RMSNorm, MLP, residual, tokenizer, or optimizer sharding, and cannot support a perplexity or stock-Llama convergence claim.

### 6.11 Aggressive pure-FP4 projection epilogue

Table 20 isolates the final pure-FP4 epilogue change. “Dual” is the earlier unified projection that emitted the ordinary Q/K layouts and then repeated fixed-scale quantization for the two pure dQ/dK operands. “Single” uses Equations 56–57 to publish all six layouts from one code fragment.

The composed measurement in Table 21 charges projection, Q/K publication, and the pure-MX backward. The baseline performs the fixed-scale pure pack in a standalone kernel after projection; the fused route consumes the projection-native bundle directly. Prepared-backward timings remain within about 2  $\mu\text{s}$ , showing that the gain is removal of producer-side conversion and movement rather than a changed attention kernel.

**Table 20:** Unified QKV projection before and after single-quantization pure-Q/K fanout. Percentages are reductions in median time.

| Shape           | dual + BF16<br>(ms) | single + BF16<br>(ms) | reduction | dual no BF16<br>(ms) | single no BF16<br>(ms) | reduction |
|-----------------|---------------------|-----------------------|-----------|----------------------|------------------------|-----------|
| S2048/H8/K1024  | 0.107936            | 0.099072              | 8.21%     | 0.104000             | 0.096896               | 6.83%     |
| S4096/H24/K1024 | 0.446112            | 0.380832              | 14.63%    | 0.410464             | 0.352384               | 14.15%    |

**Table 21:** Projection plus pure-MX backward with standalone versus projection-epilogue Q/K publication.

| Shape           | separate pack (ms) | fused epilogue (ms) | saving ( $\mu$ s) | speedup |
|-----------------|--------------------|---------------------|-------------------|---------|
| S8192/H8/K1024  | 1.102848           | 1.033216            | 69.632            | 1.067×  |
| S4096/H24/K4096 | 1.124192           | 1.027040            | 97.152            | 1.095×  |

The no-BF16 pure projection specialization compiles at 128 registers and two barriers with zero spills; the optional-BF16 specialization uses 168 registers, two barriers, and zero spills. The downstream attention kernel is unchanged at 128 registers, sixteen barriers, 231,536 bytes of shared memory, and zero spills.

## 6.12 Upstream dO publication

The fused upstream producer publishes row-major and transposed-column MXFP4 dO plus the FP32 softmax delta in one pass. At S8192/H8, five warmups and 31 rotated samples give:

**Table 22:** Upstream dO setup: three split producers versus one fused producer.

| Route                                    | median (ms) | saving (ms) | speedup |
|------------------------------------------|-------------|-------------|---------|
| row quantizer + column quantizer + delta | 0.132352    | —           | —       |
| fused row + column + delta               | 0.068416    | 0.063936    | 1.935×  |

The four payload/scale outputs are bit-exact between routes. The fused delta has maximum absolute difference  $5.96 \times 10^{-8}$  and relative L2 error  $6.72 \times 10^{-8}$  against the FP32 reference. This is a producer-side win; it does not by itself make the standalone all-MX backward the selected route.

## 6.13 Represented probability reuse

Reusing the already-produced represented E4M3 P inside the retained backward reduces S8192/H8 from 0.892480 to 0.885696 ms, a 6.784  $\mu$ s or 0.76% saving. The outputs agree with the recompute route at cosine 0.999004, 0.998987, and 0.999844 for dQ, dK, and dV. Persisting a complete forward P cache is rejected because its S8192/H8 payload and scale pages total about 272 MiB.

A stronger three-command dV candidate reused MXFP4 P directly for dS. It measured 0.782464 versus 0.756192 ms for the retained winner and returned a non-finite dV in the comparison. After reverting its two compile-time gates, the stable mixed route again produced finite outputs. The retained endpoint therefore keeps local E4M3 P reuse and does not advertise cross-forward P caching or MXFP4 P-to-dS replay.



## 6.14 Projection-backward chain

Writing fully reduced BF16 gradients into one interleaved  $[dQ, dK, dV]$  buffer and selecting a head-contiguous GEMM split gives a smaller but consistent systems gain:

**Table 23:** Old three-output attention plus three projection GEMMs versus the integrated BF16 QKV buffer and auto-split projection dgrad.

| Shape      | old chain (ms) | integrated chain (ms) | speedup |
|------------|----------------|-----------------------|---------|
| S4096/H64  | 3.276608       | 3.142528              | 1.043×  |
| S8192/H8   | 0.844672       | 0.827136              | 1.021×  |
| S8192/H24  | 2.407808       | 2.391808              | 1.007×  |
| S16384/H64 | 24.055679      | 23.471905             | 1.025×  |

dK and dV are bit-identical to the direct-BF16 endpoint. dQ differs by at most  $9.54 \times 10^{-7}$  from reduction ordering, and the resulting dX cosine is approximately 0.999992–0.999996. This path improves materialization and GEMM geometry, but it does not capture the no-publication floor because global BF16 dQ reduction remains mandatory.

The head-pipelined consumer supersedes that original whole-row polling experiment. Table 24 uses seven warmups and 31 rotated samples. The H24 forced row is the diagnostic topology; the public H24 row is the shape-dispatched fallback. Positive delta means that the public or forced entry point is faster than attention plus cuBLAS.

**Table 24:** Projection backward after moving per-head readiness into the projection K loop and adding a non-regressing public dispatch.

| Shape     | route            | attention only<br>(ms) | attention + cuBLAS<br>(ms) | result<br>(ms) | delta<br>( $\mu$ s) |
|-----------|------------------|------------------------|----------------------------|----------------|---------------------|
| S8192/H8  | public pipelined | 0.746528               | 0.774944                   | 0.772352       | +2.592              |
| S4096/H24 | public fallback  | 0.628768               | 0.734336                   | 0.733728       | +0.608              |
| S4096/H24 | forced pipelined | 0.633952               | 0.740384                   | 0.748256       | -7.872              |

At H8, moving the wait to the head boundary, adding the async-proxy publication fence, and alternating two TMEM accumulators turns the old 14–17  $\mu$ s loss into a 2.6  $\mu$ s win. One representative invocation had dX cosine 0.99999535 and relative L2 0.003051; dK and dV remain exact. At H24, a four-row-block persistent prefix plus a disjoint cuBLAS suffix reduces the old 93.6  $\mu$ s loss to 7.9  $\mu$ s, but does not win. H16 and H64 similarly favor the ordinary vendor-GEMM chain. The public entry point consequently retains the pipelined topology only for H8 and shape-dispatches wider heads to the non-regressing BF16-dQ-plus-cuBLAS path. On the final H24 check that fallback was bit-exact to the comparison output and 0.608  $\mu$ s faster, which is timing noise rather than a claimed kernel gain. The forced experimental route is available with `TK_FA4_DQ_PROJECTION_FORCE_PIPELINED`.

An isolated fixed-dQ test makes the remaining accuracy boundary explicit: the persistent projection consumer is bit-exact against `torch.mm`. Run-to-run variation originates in the nondeterministic BF16 TMA reduction order of the transposed attention grid, not in the projection MMA. Most single-run cosine values are above 0.9999, but the S8192/H8 repeat screen saw one 0.99863 cosine / 0.0523 relative-L2 excursion; this is another reason not to extend the topology to wider heads without changing the reduction owner.

### 6.15 Compact dQ projection format screen

Table 25 compares tensor formats for the isolated  $dQW_Q$  projection. The GEMM columns use already-prepared operands; the charged columns include activation conversion. Weights remain persistent. “NV delayed” uses the device-resident  $\bar{g}_D$  in Equation 87 and therefore skips a matrix-wide amax.

**Table 25:** dQ projection format timing. All entries are median milliseconds.

| Shape     | BF16     | FP8      | NVFP4    | MXFP4    | FP8 charged | NV delayed | MX charged |
|-----------|----------|----------|----------|----------|-------------|------------|------------|
| S8192/H8  | 0.078816 | 0.107776 | 0.045920 | 0.098016 | 0.251392    | 0.061952   | 0.269568   |
| S4096/H24 | 0.155968 | 0.144288 | 0.072192 | 0.125184 | 0.384608    | 0.092192   | 0.306496   |

The same-step full NVFP4 conversion plus GEMM is 0.070208 and 0.101920 ms on the two rows. Projection cosine against BF16 is 0.999301/0.999296 for FP8, 0.991005/0.990976 for NVFP4, and 0.983365/0.983188 for MXFP4. NVFP4 is the only FP4 family in this screen that repays its activation conversion while retaining approximately 0.991 cosine.

The composed chain in Table 26 includes attention, private BF16 dQ reduction, delayed-scale NVFP4 packing, and the projection. dK and dV are bit-exact with the BF16-chain outputs.

**Table 26:** Adaptive backward plus dQ projection. Speedup is BF16 chain divided by the compact NVFP4 chain.

| Shape     | attention only | BF16 chain | compact NVFP4 chain | speedup / dX cosine |
|-----------|----------------|------------|---------------------|---------------------|
| S8192/H8  | 0.750848       | 0.777568   | 0.792736            | 0.981× / 0.991005   |
| S4096/H24 | 0.630816       | 0.731968   | 0.705760            | 1.037× / 0.990976   |

The H24 projection is wide enough for NVFP4 throughput to dominate and gains 3.7%. H8 loses 1.9% because the smaller projection does not amortize the extra launch, synchronization, and packing boundary. A systems dispatcher should therefore use NVFP4 for the wide dQ projection and retain the existing BF16 projection path for the shallow H8 case. Quantizing a completed private BF16 tensor is a measured intermediate rung; only tile-ready producer fusion can remove the remaining round trip.

### 6.16 Lifetime-safe adaptive mixed dP

Separating the fixed mixed-dP scale word from the adaptive score scale page removes the earlier lifetime violation without increasing the 128-register footprint. A two-seed, three-regime screen gives:

**Table 27:** Adaptive FP8 dP/dV versus lifetime-safe adaptive mixed dP. Timing ranges span two seeds; component-cosine ranges combine dQ and dK.

| Shape     | adaptive FP8 (ms) | adaptive mixed (ms) | mixed delta     | calibrated    | unit Q/K      | sparse outlier |
|-----------|-------------------|---------------------|-----------------|---------------|---------------|----------------|
| S4096/H24 | 0.6589–0.6603     | 0.6420–0.6452       | 2.3–2.7% faster | 0.9833–0.9838 | 0.9605–0.9620 | 0.9540–0.9552  |
| S8192/H8  | 0.7670–0.7709     | 0.7770–0.7813       | 1.3–1.7% slower | 0.9827–0.9842 | 0.9588–0.9656 | 0.9547–0.9554  |

The adaptive FP8 route remains near 0.9914 calibrated dQ/dK cosine, about 0.978 in the unit-scale regime, and about 0.963 for sparse outliers. Adaptive mixed therefore restores correctness and gives

a real H24 speed point, but it does not preserve the retained route’s component accuracy and it regresses at H8. The public API exposes it explicitly as `route="mixed"`; it is not the automatic default.

### 6.17 Faster fixed mixed route

The fixed-scale mixed E4M3/MXFP4 dP route remains a useful Pareto point. On the eight-shape broad matrix it is 13.3–15.6% faster than BF16 and 3.3–5.3% faster than the higher-accuracy FP4+FP8 winner. Its calibrated dQ/dK cosine is only 0.9832–0.9840, versus 0.9912–0.9915 for the winner. The lifetime-safe adaptive specialization reaches essentially the same component-accuracy class. It is useful at H24, but the fixed route remains faster and the retained all-E4M3 dP/dV route remains more accurate.

### 6.18 NVFP4/MXFP4 pure-format screen

The pure route was also rebuilt with K16/E4M3 NVFP4 scale pages for dS, the compact Q/K operands used by dQ/dK, or both. Table 28 reports the two representative shapes. “MX/MX” is the restored pure baseline; each other row is paired with the BF16 timing from its own binary.

**Table 28:** Pure-FP4 scale-family combinations. Format order is dS / compact Q/K. Times are milliseconds.

| Shape     | formats | BF16     | candidate | dQ cosine  | dK cosine  | dV cosine |
|-----------|---------|----------|-----------|------------|------------|-----------|
| S8192/H8  | MX/MX   | 0.867200 | 0.779232  | 0.881413   | 0.882029   | 0.985053  |
|           | MX/NV   | 0.875680 | 0.802656  | 0.858788   | 0.857532   | 0.985056  |
|           | NV/MX   | 0.868640 | 0.904320  | 0.969496   | 0.969156   | 0.985056  |
|           | NV/NV   | 0.870624 | 0.979264  | non-finite | non-finite | 0.985056  |
| S4096/H24 | MX/MX   | 0.707072 | 0.635040  | 0.864334   | 0.864188   | 0.984754  |
|           | MX/NV   | 0.706368 | 0.645248  | 0.844903   | 0.846104   | 0.984521  |
|           | NV/MX   | 0.707712 | 0.735040  | 0.969362   | 0.969550   | 0.984521  |
|           | NV/NV   | 0.708480 | 0.783168  | non-finite | non-finite | 0.984521  |

NVFP4 dS is an informative accuracy rung: keeping MXFP4 Q/K raises dQ/dK cosine to approximately 0.969. Its K16 scale production and movement make it 3.9–4.1% slower than BF16, however. NVFP4 Q/K is slower and less accurate than MXFP4 Q/K in the current layout; enabling both NVFP4 changes is non-finite and 10.5–12.5% slower than BF16. All three experimental gates are restored to zero. No NVFP4/MXFP4 combination displaces the hybrid frontier.

### 6.19 No complete CuTe native-FP4 comparison

The local CuTe native-FP4 dK path does not complete in this checkout. The runnable dQ-only and BF16 controls have different precision and completeness contracts. Accordingly, this report does not quote a native-FP4 ThunderKittens/CuTe speed ratio; doing so would compare a complete kernel with an incomplete lower bound.

## 7 Profile and Remaining Ceiling

## 7.1 Latest dense-kernel profile

The latest source-correlated S8192/H8 profile excludes operand preprocessing and measures the direct-BF16 dQ specialization. Table 29 shows that the kernel is dependency-limited rather than saturated on tensor math or HBM bandwidth.

**Table 29:** S8192/H8 dense-kernel profile after BF16 dQ reduction.

| Metric                                      | Value            |
|---------------------------------------------|------------------|
| dense-kernel duration                       | 691.936 $\mu$ s  |
| tensor-pipe active                          | 21.04%           |
| issue active                                | 25.61%           |
| eligible warps per active cycle             | 0.43             |
| long-scoreboard stalls / issue-active cycle | 5.64             |
| barrier stalls / issue-active cycle         | 1.45             |
| tensor instructions                         | 211,200          |
| DRAM traffic                                | 94,448,640 bytes |
| registers / thread                          | 128              |
| dynamic shared memory                       | 231,484 bytes    |
| spill/local stack                           | zero             |

The largest source-attributed long-scoreboard samples are:

**Table 30:** Largest sampled long-scoreboard dependencies.

| Dependency                             | samples | share |
|----------------------------------------|---------|-------|
| compute-side <b>score_issued</b> wait  | 3,025   | 36.8% |
| Q-source readiness, two sites          | 744     | 9.1%  |
| statistics-side <b>score_done</b> wait | 463     | 5.6%  |
| <b>dp_done</b> wait                    | 226     | 2.7%  |
| dQ TMA-store drain/publication         | 215     | 2.6%  |

The score sample does not mean a missing barrier. It means the consumer has reached the earliest legal point and the data-producing tensor work is not yet available. Earlier experiments that rearranged score fanout moved profile samples without moving wall time; downstream  $dP/dS$  and dQ dependencies then became exposed.

## 7.2 Why tensor utilization is not higher

Backward has more tensor products than forward:

$$QK^T, \quad GV^T, \quad P^T G, \quad dSK, \quad dS^T Q. \quad (94)$$

More matrix work does not imply that these products can run concurrently. Within one tile, the dependency graph contains

$$QK^T \rightarrow P \rightarrow (dV, dS), \quad GV^T \rightarrow dS \rightarrow (dQ, dK). \quad (95)$$

The tensor issuer therefore alternates between products whose operands are created by scalar/packed producer work. Score has to fan out to P; dP has to be centered; dS has to be quantized in two orientations; and dQ eventually has to leave TMEM through a cross-owner reduction.

Forward can devote most TMEM to two score/output pipelines. Backward keeps 320 columns permanently occupied by dK and dV, leaving one transient score/dP/dQ region. Thus the forward strategy of adding an independent score accumulator is not a local template change: it would evict live gradient state. The three exact K64 score instructions are also a hard  $D = 192$  requirement. Dropping one changes the represented rank and sharply degrades accuracy.

### 7.3 What the projection result changes

Before the native learned projection, the low-precision backward gain could be lost to Q/K reduction, packing, transposition, and HBM round trips. The new producer removes that systems bottleneck for sufficiently deep projection K. It therefore raises the meaningful optimization ceiling: improvements inside attention are no longer automatically canceled by fake quantization outside it.

The remaining boundaries are now:

1. **Scale availability.** The fused Q/K epilogue consumes  $m_h$ ; producing Equation 13 without a BF16 Q/K round trip still requires an integration policy.
2. **Score production and fanout.** Exact D192 needs three K64 commands and one completed represented score before P.
3. **dQ reduction topology.** K/V ownership creates many partial dQ contributors. BF16 halves traffic, but publication remains.
4. **Projection-backward consumer.** Per-head readiness and two TMEM accumulators reach a small H8 win, but the consumer still observes a globally reduced BF16 tile. Its bounded grid leaves an exposed tail for wider output matrices, where shape dispatch returns to cuBLAS.

### 7.4 Quantified ceilings

At S8192/H8:

$$T_{\text{direct BF16 endpoint}} \approx 0.746 \text{ ms}, \quad (96)$$

$$T_{\text{dense kernel profile}} \approx 0.692 \text{ ms}, \quad (97)$$

$$T_{\text{no dQ publication diagnostic}} \approx 0.693 \text{ ms}. \quad (98)$$

The near equality of the latter two numbers is coincidental measurement context, but both place the remaining standalone opportunity near  $53 \mu\text{s}$ , or roughly 7% of the direct-BF16 endpoint. A real fused consumer must spend some of that budget on reduction and projection work, so 0.693 ms is an optimistic floor rather than a forecast.

### 7.5 Plausible breakthrough directions

The profile and ownership proof narrow the next high-leverage work:

1. **Projection-native adaptive scales.** Emit per-head scale statistics with the learned projection, use a delayed/calibrated scale, or train a scale predictor so the four E2M1 layouts never depend on a BF16 sidecar.
2. **Q-centric reduction and projection dgrad.** Make one cluster own a complete dQ tile, or fuse a hierarchical owner reduction into Equation 73. The new head-pipelined consumer moves readiness into the K loop, but it does not eliminate the BF16 scratch; eliminating that scratch is the path to the no-publication floor.
3. **Model-supported lower score rank.** Two K64 score commands become legitimate only if the learned Q/K projection is trained to produce a real rank-128 subspace. A fixed post-hoc projection is not enough.
4. **Upstream low-precision gradient operands.** Emit reusable E4M3 or MXFP4 V/G views from adjacent model kernels, preserving the fixed-power contracts without standalone preprocessing.

More barrier state, scalar atomics, or another TMEM rearrangement within the current 512-column map has a much smaller measured ceiling.

## 8 Discussion and Merge Boundary

### 8.1 What the current result establishes

The retained route demonstrates that low-precision backward should be treated as a represented operator, not as BF16 backward followed by casts. The fixed powers in Equations 21–27 remove general rescaling from the hot path; P and dS are quantized at production; and adaptive Q/K metadata is consumed by tensor scale pages and gradient epilogues without adding an MMA.

The learned-projection result adds the systems half of that argument. NVFP4 tensor work is useful only if its operand preparation and output layouts are part of surrounding producers. The unified epilogue emits one Q/K payload for both directions, transposed MXFP4 V, and optional BF16 Q/K/V. It thereby avoids multiple BF16 rereads, standalone transposes, and fake-quantization launches, producing a much larger systems gain than another local backward barrier adjustment.

The final pure-FP4 specialization applies that rule more aggressively. It quantizes each Q/K projection fragment once at the fixed pure scale and fans the same codes into all six consumer layouts. This removes 69.6–97.2  $\mu$ s from two projection-plus-backward chains. The result also separates the remaining limits cleanly: pure operand generation is no longer the main setup penalty, yet the mixed-dP hybrid remains faster because block maxima, dual dS scale publication, and score fanout are exposed inside backward. Changing those internal pages from MXFP4 K32/E8M0 to NVFP4 K16/E4M3 does not recover the gap. The more accurate NVFP4-dS variant is slower than BF16, and the combined NVFP4 diagnostic is non-finite.

The lifetime-safe adaptive mixed-dP result is a narrower positive outcome. Once the dP tail word and adaptive score page have separate lifetimes, the route is stable at the same zero-spill register limit. It wins at H24 but not H8 and remains in the approximately 0.983 dQ/dK accuracy class. This confirms that adaptive scaling and mixed operand precision are independent choices; fixing the former does not erase error introduced by the latter.

The upstream dO result supports the same rule from the other side of the graph. Row and column MXFP4 representations and the softmax delta should be published by one producer, not reconstructed through three kernels immediately before backward. This halves that setup boundary without changing any backward arithmetic.

The composed MFU result tests the rule at a model-facing boundary. Keeping the FP4 FA4 forward and FP4+FP8 backward together raises useful algorithmic MFU on all five measured shapes. The gain is modest at H16 and largest at wide H64 projection widths, where projection-native NVFP4 work amortizes packing and publication. The faster activation-only stacked handoff is not silently used for full training: the latter retains BF16 QKV publication so the QKV weight-gradient GEMM remains present.

## 8.2 Why backward does not reproduce the forward gain directly

The companion forward report uses two matrix products and two major TMEM accumulators, then attacks the serial score-to-P publication path. Backward has five products, persistent dK/dV state, two dS orientations, and a cross-owner dQ reduction. Its critical path is therefore not just exponential evaluation. Approximating  $2^x$  more aggressively cannot remove the score TMEM fanout, dP centering, dS publication, or dQ reduction.

The current backward also chooses E4M3 P for its accuracy-oriented route, whereas the forward report's retained full-FP4 policy uses MXFP4 P/V and a direct E2M1 code map. Those are different represented operators. A local represented-P replay saves 0.76%, but materializing forward P for backward would write and reread 272 MiB at S8192/H8. The profitable analogue of forward reuse must therefore remain on chip or change the consumer topology; an HBM cache is not a useful bridge.

## 8.3 What the six integration tests rule out

The tests narrow the next breakthrough substantially. It is not a second Q/K format, a standalone V quantizer, three separate dO setup kernels, a global P cache, or another barrier. Per-head polling inside the projection K loop is enough for a small H8 win, and double-buffering the projection's private TMEM removes its MMA-epilogue bubble. It is not enough at wider heads. The remaining high-value boundary is the global dQ reduction: projection backward must consume the reduction tile before, or instead of, BF16 global materialization. Observing the tile immediately after publication still pays the global rendezvous and leaves a bounded-consumer tail.

The compact NVFP4 final-tile handoff sharpens this boundary. Its isolated projection, including delayed-scale conversion, beats BF16 at both measured widths and reaches approximately 0.991 cosine. The composed chain wins 3.7% at H24 but loses 1.9% at H8 because it still waits for and rereads the private BF16 reduction. Format choice can accelerate the consumer; it cannot by itself remove the producer topology that defines the ceiling.

The implemented two-lane hierarchy further rules out a weaker interpretation of that target. Projection can consume two BF16 owner sums directly and avoid ever constructing completed dQ, but it is slower because BF16 partial traffic is unchanged and the consumer must duplicate the weight MMA or fold the lanes. The useful breakthrough must therefore combine projection integration with pre-publication owner grouping or a compact FP8/FP4 partial representation. Moving the final addition alone does not approach the no-publication timing diagnostic.



The stacked-NVFP4 version reaches the same conclusion through a different consumer. Folding two dQ lanes, inverse RoPE, and local quantization in one K128 producer is accurate, but its readiness signals perturb the attention schedule enough to lose. Removing those signals makes the one-lane stacked layout a small activation-only winner, but also restores completed BF16 dQ. It is a useful publication optimization, not the sought reduction-topology breakthrough.

Storage reclamation also has a hard hardware boundary. Projection scratch can be unioned because Q/K code staging and V BF16-pair staging have disjoint lifetimes. The backward mixed-dP row cannot simply shrink from 128 to 96 bytes: the grouped SM100 depth transaction does not retire with that geometry. The final 32 bytes are logically padding but are not currently reclaimable by a legal drop-in descriptor.

## 8.4 Accuracy and training limits

The reported cosine values cover synthetic calibrated, unit-scale, and sparse-outlier regimes. The new 500-step teacher/student experiment adds a block-level optimization trajectory with refreshed weights and complete projection weight gradients. Across three seeds, low precision retains 89.7–99.6% of BF16’s absolute validation-loss improvement. This does not establish language-model convergence, final perplexity, optimizer-state stability over many layers, or sensitivity to realistic activation tails. The learned NVFP4 projection itself has Q/K/V cosine near 0.991, while attention backward has a separate approximately 0.991 dQ/dK cosine. Native forward output cosine is 0.9907–0.9913 on the two unified-projection checks. The compact NVFP4 dQ projection contribution has approximately 0.991 cosine against the BF16 projection; FP8 is approximately 0.9993 and MXFP4 approximately 0.9832–0.9834 on the same boundary. Their composed error is not the product of those scalars. The trajectory experiment shows that reducible error falls, but the FP4 forward contributes a native teacher floor of 0.093–0.098 relative MSE and the final shared-BF16 evaluation remains 25.5% above the BF16 student’s loss on average. The raw loss gap must remain visible beside the floor-adjusted convergence rate.

Robust per-head clipping deliberately sacrifices extreme values to protect the bulk distribution. Per-token or blockwise Q/K scales could retain more outliers, but scales varying along the  $dQ/dK$  reduction dimension require additional instruction scale pages and TMEM movement. Quantization-aware training is the natural larger validation framework [9].

## 8.5 Hardware and shape scope

The active specialization is causal, batch one, sequence divisible by 128,  $D_{QK} = 192$ , and  $D_V = 128$  on data-center Blackwell. It supports the measured S4096, S8192, and S16384 families and selected head counts. The projection kernel requires matrix dimensions aligned to its NVFP4 tiling. GQA, variable-length batches, dropout, arbitrary head dimensions, and a portable non-TMEM implementation are outside this report.

The 512-column alias schedule, `tcgen05` scale pages, and two-CTA matrix shapes are SM100-specific [6]. A device with another score bank, different scaled-MMA K granularity, or cheaper cross-cluster reduction could choose a different ownership plan.

## 8.6 What belongs to prior structure

FlashAttention supplies the recomputation and IO-aware backward formulation [1]. FA4 supplies the hardware-asymmetric, warp-specialized scheduling framework [8]. ThunderKittens supplies the optimized V382 control and the tile abstractions [3]. HAO’s forward FP4 work is a structural reference for block-scaled tensor operations, TMEM lifetime reuse, and producer-side quantization [2]. This report’s contribution is the backward precision factorization, adaptive Q/K contract, retained TMEM handoffs, BF16 dQ endpoint, unified learned-QKV publication, fused upstream dO contract, and measured projection-gradient boundary built on those foundations.

## 8.7 Should this merge with the forward report?

Not yet as a single method section. The reports should remain separate while the following are unresolved:

- the forward and backward P representations differ, and the measured HBM P cache is not viable;
- projection-scale production is still an explicit integration contract;
- composed forward-plus-backward accuracy and a synthetic block-level trajectory are measured, but stock-model loss and perplexity are not;
- the pure-FP4 backward route is validated but experimental, while the retained backward is the more accurate FP4+FP8 route;
- the fused dO producer and all-MX consumer compose successfully, but that endpoint is not the selected performance/accuracy frontier.

A later umbrella manuscript should merge the shared introduction, Blackwell format background, notation, and end-to-end evaluation. It should keep separate forward and backward architecture/method sections, then add one systems section for projection epilogues and one composed training-results section. This avoids duplicating background without implying that the two kernels currently implement the same probability operator.

## 9 Conclusion

The retained Blackwell backward uses adaptive FP4 Q/K for score and  $dQ/dK$ , E4M3 operands for  $dP/dV$ , direct E4M3 dS production, FP32 TMEM accumulation, and BF16 dQ reduction. Its represented equations are explicit: P is scaled by  $2^8$ ,  $dP - r$  by  $2^4$ , and their product lands directly at the  $2^{12}$  dS scale consumed by mixed E4M3×E2M1 tensor operations. This route is 11.3–16.4% faster than the copied BF16 control on the measured long-sequence matrix while preserving approximately 0.9915 dQ/dK cosine in the calibrated regime.

The native NVFP4 learned QKV projection is the larger systems result. It emits one forward/backward Q/K payload, exact backward views, transposed MXFP4 V, and optional BF16 Q/K/V from its output fragments. On the measured projection shapes it is 1.41–1.68× faster than three BF16 GEMMs plus Q/K packing with BF16 publication, and 1.60–1.93× faster without it. Pair-native

RoPE is fused before every Q/K publication and saves  $16.128 \mu\text{s}$  against a strong standalone forward transform at S4096/H24/K3072. On the final five-shape matrix that keeps FP4 FA4 forward in every low-precision row, the RoPE-aware dense decoder boundary is  $1.118\text{--}1.720\times$  faster than BF16 for activation gradients and  $1.096\text{--}1.469\times$  when projection weight gradients are charged. At S4096/H24/K3072, full-training algorithmic MFU rises from 26.94% to 32.38% on a common 2.5-PFLOP/s dense denominator. This boundary remains a dense  $D_{QK} = 192$  result, not yet a stock grouped-query Llama drop-in. The fused dO row/column/delta producer adds a separate  $1.935\times$  setup win.

The final stacked QKV-gradient producer fuses inverse RoPE, K16 scale selection, and NVFP4 packing before one dX projection. With the normal unsigned dQ endpoint it improves the S4096/H24 activation boundary from 1.578336 to 1.566048 ms; a genuinely hierarchical two-lane producer regresses to 1.635936 ms because its readiness signaling slows attention. The stacked winner is therefore activation-only. Full training keeps the interleaved BF16 QKV gradient needed for weight gradients.

A 500-step, three-seed teacher/student proxy provides the first optimization trajectory rather than another one-step cosine. Low precision retains 93.2% of BF16's absolute validation-loss improvement on average and reduces its floor-adjusted excess error at least as quickly in all three runs. Raw native loss is still higher because the FP4 forward has a 0.093–0.098 relative-MSE teacher floor, and a common BF16 evaluator leaves final loss 25.5% higher on average. These data support similar block-level optimization progress, not equal final loss or stock-Llama language-model convergence.

For the aggressive pure-FP4 endpoint, the projection epilogue now performs a single fixed-scale Q/K quantization and publishes all six required backward layouts directly. This reduces unified projection time by 6.8–14.6% and the measured projection-plus-pure-backward chains by 6.7–9.5%. The pure kernel is 10.0–11.4% faster than BF16 on the seven-shape sweep, but remains slower than the mixed-dP hybrid and its dQ/dK cosine is only 0.864–0.887. It is therefore a completed aggressive endpoint, not the new default. Substituting NVFP4 K16 scales does not rescue it: NVFP4 dS improves component cosine to approximately 0.969 only by becoming slower than BF16, NVFP4 Q/K regresses both speed and accuracy, and the combined diagnostic is non-finite. Pure FP4 is therefore not a competitive backward default in the current SM100 ownership and scale-publication topology.

The repaired adaptive mixed-dP route is a valid intermediate Pareto point. It is 2.3–2.7% faster than adaptive FP8 dP/dV at S4096/H24, but 1.3–1.7% slower at S8192/H8 and lowers calibrated dQ/dK cosine from approximately 0.991 to 0.983. It is exposed explicitly rather than selected automatically.

The integration experiments also establish the present ceiling. Local represented-P reuse is worth only 0.76%; an HBM P cache is too large; and the 96-byte mixed-dP transport is not a legal completing transaction. Moving per-head dQ readiness into the projection K loop, ordering the async proxy, and alternating two private TMEM accumulators improves the original direct consumer enough to beat attention plus cuBLAS by  $2.6 \mu\text{s}$  at S8192/H8. The same bounded topology remains  $7.9 \mu\text{s}$  behind at S4096/H24 even after a vendor-GEMM tail, so wider heads shape-dispatch to the non-regressing BF16 dQ plus cuBLAS chain. The next breakthrough is therefore not another barrier rearrangement. It is projection-native adaptive scale production and a reduction topology that lets projection backward consume dQ before a standalone global BF16 tensor exists. A direct two-lane implementation now proves that topology algebraically and numerically, but remains  $32.1 \mu\text{s}$  behind at S8192/H8 because it preserves BF16 partial traffic and duplicates the projection product. The

remaining requirement is stricter: group owners on chip before publication, or publish compact FP8/FP4 partials whose repeated products are paid for by low-precision tensor throughput.

The compact NVFP4 final-tile experiment validates the consumer half of that direction. With delayed device-resident scale state, NVFP4 conversion plus projection reaches approximately 0.991 cosine and the composed H24 chain is 3.7% faster than BF16; H8 is 1.9% slower. It still quantizes completed private BF16 dQ scratch. The next implementation must move that quantizer to tile readiness or an on-chip super-owner so the low-precision handoff removes, rather than follows, the BF16 reduction round trip. The analogous RoPE opportunity has now been tested. Fusing  $R^T dQ$  and  $R^T dK$  into the delayed-scale NVFP4 quantizer is bit-exact, but saves only 2.720  $\mu s$  without BF16 republication and 3.392  $\mu s$  with it. At S4096/H24/K4096 those component savings become 0.46% for activation gradients and 0.60% when both projection weight-gradient GEMMs are charged. The row-owned quantizer cannot make the 0.115-ms transform free. A forward-sized systems gain requires inverse RoPE inside a projection consumer that receives the completed reduction tile before global BF16 publication.

## References

- [1] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. *Advances in Neural Information Processing Systems*, 35, 2022. URL <https://arxiv.org/abs/2205.14135>.
- [2] HAO AI Lab. Fp4 flash attention 4 on blackwell. GitHub repository, branch `fp4`, commit `9b0abef`, 2026. URL [https://github.com/hao-ai-lab/flash-attention-fp4/tree/fp4/flash\\_attn/cute](https://github.com/hao-ai-lab/flash-attention-fp4/tree/fp4/flash_attn/cute).
- [3] Hazy Research. Thunderkittens: Simple, fast, and adorable ai kernels, 2024. URL <https://github.com/HazyResearch/ThunderKittens>.
- [4] NVIDIA Corporation. Introducing nvfp4 for efficient and accurate low-precision inference, 2025. URL <https://developer.nvidia.com/blog/introducing-nvfp4-for-efficient-and-accurate-low-precision-inference/>.
- [5] NVIDIA Corporation. Nvidia gb200 nvl72 specifications, 2026. URL <https://www.nvidia.com/en-gb/data-center/gb200-nvl72/>. The published superchip contains two GPUs; dense throughput is one-half the listed sparse specification.
- [6] NVIDIA Corporation. Parallel thread execution isa, version 8.8, 2026. URL <https://docs.nvidia.com/cuda/parallel-thread-execution/>.
- [7] Open Compute Project. Ocp microscaling formats (mx) specification, September 2023. URL <https://www.opencompute.org/documents/ocp-microscaling-formats-mx-v1-0-spec-final-pdf>.
- [8] Ted Zadouri, Markus Hoehnerbach, Jay Shah, Timmy Liu, Vijay Thakkar, and Tri Dao. Flashattention-4: Algorithm and kernel pipelining co-design for asymmetric hardware scaling, 2026. URL <https://arxiv.org/abs/2603.05451>.
- [9] Peiyuan Zhang, Matthew Noto, Wenxuan Tan, Chengquan Jiang, Will Lin, Wei Zhou, and Hao Zhang. Attn-qat: 4-bit attention with quantization-aware training, 2026. URL <https://arxiv.org/abs/2603.00040>.

## A Precision Ladder

### A.1 Implemented routes

The copied scheduler contains a ladder of modes so format changes can be isolated without changing the BF16 control. Table 31 states the mathematical intent rather than every historical compile-time probe.

**Table 31:** Implemented backward precision ladder.

| Mode | Q/K score     | $dP, dV$ operands            | $dS, dQ, dK$                      | Status                     |
|------|---------------|------------------------------|-----------------------------------|----------------------------|
| 0    | BF16          | BF16                         | BF16                              | copied V382 control        |
| 1    | E4M3          | BF16                         | E4M3 dS, E4M3 $dQ/dK$             | native-FP8 diagnostic      |
| 2    | E2M1          | BF16                         | E4M3 dS, mixed E4M3 $\times$ E2M1 | first FP4 rung             |
| 5    | adaptive E2M1 | E4M3 $dP, dV$                | E4M3 dS, mixed dQ/dK              | retained winner            |
| 12   | E2M1          | mixed E4M3/MXFP4 dP; E4M3 dV | E4M3 dS, mixed dQ/dK              | fixed-scale speed frontier |
| 12+A | adaptive E2M1 | mixed E4M3/MXFP4 dP; E4M3 dV | E4M3 dS, mixed dQ/dK              | lifetime-safe H24 frontier |
| 11   | E2M1 NVFP4    | MXFP4 dP and dV              | MXFP4 dS and scaled MXFP4 dQ/dK   | experimental pure route    |

### A.2 Fixed mixed-dP equation

The faster mode-12 route keeps the first K64 of  $V$  and  $G$  in E4M3 and packs the second K64 as fixed-scale E2M1. Abstractly,

$$D_P^{\text{mixed}} = \text{MMA}_{\text{E4M3}}(\bar{G}_{0:64}, \bar{V}_{0:64}^T) + \gamma \text{MMA}_{\text{E2M1}}(\bar{G}_{64:128}, \bar{V}_{64:128}^T), \quad (99)$$

where  $\gamma$  is folded into a reused fixed scale page. dV remains the all-E4M3 operation in Equation 24. A reusable  $V$  row contains 64 E4M3 bytes, 32 packed E2M1 bytes, and 32 padding bytes. Forward can emit this row once so backward does not repay the  $V$  conversion.

This route is 3.3–5.3% faster than the retained winner on the broad matrix, but its calibrated dQ/dK cosine is approximately 0.983 rather than 0.991. The scale is fixed and distribution-dependent.

Mode 12+A supplies the robust per-head Q/K record to the same mixed-dP operator. Its score and dQ/dK scale page is distinct from the fixed dP tail word, resolving the lifetime bug in the original attempt. It is 2.3–2.7% faster than adaptive all-E4M3 dP/dV at S4096/H24 but 1.3–1.7% slower at S8192/H8. Because mixed dP still controls its component error, calibrated dQ/dK cosine remains approximately 0.983.

### A.3 MXFP4 block representation

For a 32-value block  $B$  with magnitude  $a_B$ , choose a power-of-two scale

$$s_B = 2^{e_B}, \quad e_B \in \{\lfloor \log_2(a_B/6) \rfloor, \lceil \log_2(a_B/6) \rceil\}, \quad (100)$$

subject to E8M0 range and the selected rounding policy. The represented block is

$$\bar{X}_B = Q_{E2M1}(X_B/s_B), \quad \hat{X}_B = s_B \bar{X}_B. \quad (101)$$

The tensor instruction receives swizzled  $32 \times 16$  scale pages in TMEM.

#### A.4 Experimental pure-MXFP4 backward

The pure route applies Equation 101 to V, G, P, and dS. At the represented-operator level:

$$D_P^{(4)} = \text{MMA}_{MX4}(\bar{G}_4, \bar{V}_4^\top; s_G, s_V), \quad (102)$$

$$\widetilde{dV}^{(4)} = \text{MMA}_{MX4}(\bar{P}_4^\top, \bar{G}_4; s_P, s_G). \quad (103)$$

The raw MXFP4 dP accumulator carries the implementation’s standard width-six reconstruction factor:

$$\widetilde{dP}^{(4)} = \frac{1}{36} D_P^{(4)}. \quad (104)$$

The producer folds  $1/36$  into centering instead of materializing a corrected dP tile:

$$dS^* = \hat{P}_4 \odot \left( \frac{1}{36} D_P^{(4)} - r \mathbf{1}^\top \right). \quad (105)$$

It then emits two independently scaled MXFP4 orientations,

$$(\bar{dS}_K, s_{dS,K}) = Q_{MX4}(dS^*) \quad \text{for } dK, \quad (106)$$

$$(\bar{dS}_Q, s_{dS,Q}) = Q_{MX4}((dS^*)^\top) \quad \text{for } dQ, \quad (107)$$

and consumes compact Q/K operands:

$$\widetilde{dQ}^{(4)} = \alpha \text{MMA}_{MX4}(\bar{dS}_Q, \bar{K}_4; s_{dS,Q}, s_K), \quad (108)$$

$$\widetilde{dK}^{(4)} = \alpha \text{MMA}_{MX4}(\bar{dS}_K, \bar{Q}_4; s_{dS,K}, s_Q). \quad (109)$$

P uses a forward-style log-domain producer. For one probability block,

$$x_{ij} = (\tilde{Z}_{ij} - L_i) \log_2 e - e_B + \log_2 6, \quad (110)$$

$$\bar{P}_{4,ij} = Q_{E2M1}(2^{x_{ij}}), \quad \hat{P}_{4,ij} = \frac{s_B}{6} \bar{P}_{4,ij}. \quad (111)$$

The block maximum that selects  $e_B$  is shared with quantization, avoiding a second full probability pass.

The initial 148-tile ceiling validated every active matrix output bit-exactly against products of the captured represented operands. Its recommended NV-dP/MX-dS stage-6 point measured  $130.142 \mu s$  for 148 independent tiles. The completed long-sequence mode 11 now also consumes projection-native Q/K/V and producer-native dO payloads. Table 32 reports the prepared-kernel comparison from one fixed-seed calibrated sweep.

Pure MXFP4 is 10.0–11.4% faster than BF16 in this matrix and beats the ordinary fixed FP4+FP8 route on the two S4096 rows. It does not beat the mixed-dP route on any row, and it loses to adaptive FP4+FP8 once sequence is 8192 or longer. More importantly, its dQ/dK component cosine is only about 0.864–0.887, while aggregate cosine near 0.985 is dominated by the more accurate dV component. The stable long-sequence accuracy-oriented API therefore remains mode 5; mode 11 is a validated aggressive endpoint.



**Table 32:** Prepared backward precision frontier. Times are median milliseconds; the final column is aggregate dQ/dK/dV cosine of pure MXFP4 versus BF16.

| Shape      | BF16      | FP4+FP8   | mixed dP  | adaptive FP4+FP8 | pure MXFP4 | pure cosine |
|------------|-----------|-----------|-----------|------------------|------------|-------------|
| S4096/H24  | 0.707936  | 0.647104  | 0.613664  | 0.660096         | 0.637120   | 0.984797    |
| S4096/H64  | 1.655648  | 1.513952  | 1.432832  | 1.539360         | 1.482304   | 0.984614    |
| S8192/H8   | 0.869984  | 0.758656  | 0.745664  | 0.773472         | 0.781920   | 0.984653    |
| S8192/H24  | 2.212064  | 1.923200  | 1.885216  | 1.956544         | 1.971136   | 0.985197    |
| S8192/H64  | 5.558176  | 4.871296  | 4.739136  | 4.940192         | 4.958176   | 0.985087    |
| S16384/H24 | 7.758656  | 6.510784  | 6.536256  | 6.610144         | 6.874848   | 0.985217    |
| S16384/H64 | 20.275393 | 17.163391 | 17.102079 | 17.419872        | 17.997663  | 0.985224    |

## A.5 Why pure FP4 is not the headline

The pure route eliminates more operand bytes but adds block maxima, E8M0 scale publication, two dS scale orientations, and additional TMEM scale lifetimes. Backward’s scalar and synchronization work is already exposed, so replacing E4M3 instructions with MXFP4 does not automatically translate the peak tensor-core ratio into wall time. Fusing Q/K quantization into projection removes 69.6–97.2  $\mu$ s from two composed checks, but it does not shorten those in-kernel maxima and scale handoffs. Its numerical contract is also more aggressive than the retained approximately 0.991 dQ/dK point. It remains the right research ladder, but not the current default.

Replacing either scale family with NVFP4 K16/E4M3 pages does not change that conclusion. NVFP4 dS with MXFP4 Q/K improves dQ/dK cosine to approximately 0.969 but becomes slower than BF16; NVFP4 Q/K loses speed and accuracy; and the combined NV/NV diagnostic is non-finite. The retained pure specialization therefore keeps MXFP4 dS and MXFP4 Q/K scale pages.

## B Rejected and Diagnostic Experiments

### B.1 Two-command D192 score

Omitting the final K64 score command changes the score contraction from rank 192 to at most rank 128. It saved only 12.7  $\mu$ s at S8192/H8 and 46.2  $\mu$ s at S16384/H24 in one calibrated screen, while unit-scale Q/K dQ/dK cosine fell to approximately 0.825. Sparse-outlier cosine was approximately 0.923–0.925. CountSketch, dense Rademacher projection, and energy-selected chunks did not recover general inputs. The mode is disabled.

### B.2 Additional score pipelines

Split N64/N96 score publication and early-prefix consumption moved stall samples but did not reduce wall time. A cross-iteration score preissue reused the existing exchange-tail scratch and failed its first runtime gate. A second complete score accumulator is impossible in the active 512-column map. Streaming half of dV to recover 64 columns lost more to scratch drain and scatter than a later score could save.



### B.3 dQ publication alternatives

Pairing twelve  $32 \times 32$  dQ reductions into six  $64 \times 32$  transactions was correct but 0.86% slower at S8192/H8 because the pairwise barriers became exposed. The equivalent mixed-route pairing was about 1.1% slower. Accumulating unscaled dQ followed by a linear scale pass added a complete dQ read/write and regressed timing and rounding.

A timing-only half-owner diagnostic bounded a four-CTA hierarchical reduction at roughly 3–4% before DSM coordination cost. Replacing TMA reduction with scalar atomics would increase instruction issue while retaining L2 serialization. No alternative was retained.

A later two-lane implementation did perform the final dQ addition in the stacked NVFP4 producer, together with inverse RoPE and QKV packing. It is correct but measures 1.635936 ms at the S4096/H24/K3072 activation boundary, versus 1.578336 ms for the selected interleaved path and 1.566048 ms for the unsigaled one-lane stacked control. Nsight Systems attributes approximately  $37 \mu\text{s}$  to enabling tile-ready signaling even for one lane and roughly  $48 \mu\text{s}$  to the actual two-lane schedule. K64 TMA descriptors and three attempts to split BF16/packed shared pages also produced illegal accesses and were discarded. The retained producer uses the proven K128 descriptor and eight warps; the two-lane mode remains an opt-in structural diagnostic.

### B.4 Aliased adaptive mixed-dP scale page

The first adaptive mixed-dP attempt reused the live score-scale TMEM word for the fixed E2M1 dP tail. Robust per-head Q/K metadata made the alias visible as repeatable tile-local dQ corruption. That exact layout remains rejected. The replacement assigns the dP word a distinct lifetime-safe page, compiles at 128 registers without spills, and is exposed explicitly as `route="mixed"`. It is not the default because it regresses H8 timing and reduces calibrated dQ/dK cosine from approximately 0.991 to 0.983.

### B.5 Whole-row tile-ready projection consumer

The first head-major grid used one readiness gate in front of the whole projection tile. It measured  $16.608 \mu\text{s}$  slower than attention plus cuBLAS at S8192/H8. Launching the polling grid first can also prevent producer admission and is invalid. This version was superseded by per-head waits inside the projection K loop and two alternating TMEM accumulators; the replacement saves  $2.592 \mu\text{s}$  at H8. The rejected result remains relevant because it shows that launch order alone is not reduction-topology integration.

### B.6 Forward probability cache and MXFP4 P-to-dS replay

Local represented-E4M3 P reuse saves  $6.784 \mu\text{s}$  at S8192/H8, but persisting the complete forward probability matrix would require approximately 272 MiB for one MXFP4 payload plus scale pages. A forward write and backward read cannot repay a single-digit-microsecond local saving, so no global cache was implemented.

A stronger candidate kept the three-command mixed dV route and reused its MXFP4 P fragment directly for dS. It regressed from 0.756192 to 0.782464 ms, reduced dQ/dK cosine to 0.981658/0.981543, and returned a non-finite dV. Its two feature gates were restored to zero and the stable route was rebuilt.

## B.7 Compact 96-byte mixed-dP transport

Each retained mixed-dP row transfers 128 bytes although its final 32 bytes are not consumed by the dP contraction. A 96-byte depth descriptor appears to reclaim the padding, but the grouped SM100 transaction does not retire under the score/dO barrier even when the tail MMA is omitted. The 128-byte row is therefore retained. The padding is a real storage inefficiency, not currently a legally removable one.

## B.8 Pure-MX production dispatch

The all-MX endpoint now passes launch, finiteness, payload, and output validation with projection-native Q/K/V and producer-native dO. It is no longer rejected as an implementation. It is rejected only as the default: the broad sweep in Table 32 leaves it behind the mixed dP route at every shape, while pure dQ/dK cosine is approximately 0.864–0.887. The route remains exposed as an aggressive research endpoint rather than being promoted over the higher-accuracy hybrid.

## B.9 Additional pure-FP4 epilogue fusion

The final pass tested dS-scale fusion, direct P-scale publication, direct dQ scale application, interleaved dual E2M1 packing, early dO reload, a deferred native-dS TMEM wait, streamed dual dPsum, and two P/dO K-block interleave orders. These variants either regressed or changed only stall attribution without improving calibrated wall time. Their compile-time gates are all restored to zero. The retained epilogue fusion is instead upstream: one Q/K projection quantization fans out into all required layouts, where it removes a complete conversion/publication boundary.

A final scale-family sweep tested NVFP4 K16/E4M3 pages for dS and compact Q/K. NVFP4 dS improved dQ/dK cosine to approximately 0.969 but was 3.9–4.1% slower than BF16. NVFP4 Q/K was slower and less accurate than MXFP4 Q/K, and the combined diagnostic returned non-finite dQ/dK. The three gates are restored to zero.

## B.10 Global and row/block adaptive Q/K scales

A single global amax overreacted to sparse outliers, giving approximately 0.753/0.757 dQ/dK cosine at S4096/H24. Robust per-head scaling fixed most of that failure. Finer row/block Q/K scales were not retained because their scale varies along the dQ/dK reduction dimension; mixed gradient instructions would need additional scale pages rather than the current scalar epilogue.

## B.11 Barrier-only rearrangements

Score, dP, and dQ issue markers were useful because they changed when TMEM operations could enter the hardware scoreboard. Additional fanout barriers, uniform rendezvous variants, and producer/consumer relays usually shifted sample attribution or exposed another wait without changing the underlying data-ready time. The latest profile therefore treats barriers as a symptom below score production, not the next primary optimization target.

## B.12 Cooperative inverse-RoPE staging in the NVFP4 quantizer

The direct fused inverse-RoPE quantizer is correct and retains a useful 122-register no-publication specialization (153 registers with BF16 republication), but its row-owned threads read the rotary tables with a 192-byte row stride. Two cooperative alternatives staged either one 32-plane half at a time or one complete 64-plane QKV column tile. Both made the table traffic coalesced and remained bit-exact. Neither was retained: the half-tile form compiled at 182/234 registers for no-publication/publication, and the full compact tile reached 218–250 registers. Their extra barriers and lower occupancy made them slower than both the direct fusion and the separate transform. Coalescing alone is not a win when it extends the lifetime of the quantizer’s complete row fragment.

## C Reproduction and Evidence

### C.1 Source boundary

The retained implementation is under `tk_fa4/lowp_fa4_bwd/`. The primary files are:

- `lowp_fa4_bwd.cu`: bindings, operand producers, dispatch, and launch contracts;
- `b300_bwd_cute16_kernel_candidate.cuh`: copied V382 schedule, low-precision modes, TMEM map, and retained handoffs;
- `projection_fp4_epilogue.cuh`: persistent NVFP4 QKV learned projection, shared Q/K publication, projection-native MXFP4 V, and the single-output generic NVFP4 projection consumer;
- `dq_projection_consumer.cuh`: head-pipelined, double-TMEM dQ projection consumer;
- `hierarchical_qkv_nvfp4_quantize.cuh`: one-/two-lane stacked QKV-gradient inverse-RoPE and delayed-scale NVFP4 producer;
- `evaluate_llama_attention_e2e.py` and `evaluate_llama_attention_convergence.py`: composed timing/MFU and multi-step trajectory harnesses;
- `interface.py`: public adaptive operand, backward, and projection APIs.

The upstream dO producer is implemented in `TK_quantisation/mxftp4_v3/tk_quantize.cu`; its default integration lives in `tk_fa4/fp4_pv_experiments.py`.

The forward report at `results/fp4_fa4_technical_report_20260728/` was used as a read-only structure and notation reference.

### C.2 Build

From `tk_fa4/lowp_fa4_bwd/`:

```
make -B -j1
```

The build used for the reported projection integration exited with status zero. The base unified no-BF16 projection specialization reports 128 registers, two barriers, 8,592 bytes of static shared

memory, and zero spills; the RoPE-enabled BF16-publication specializations report 168 registers, two barriers, the same shared-memory allocation, and zero spills. Projection-only publication reports 96 registers and zero spills; the Q/K-only fused specialization reports 120 registers. The generic single-output NVFP4 dQ consumer reports 86 registers, two barriers, and zero spills. The adaptive all-E4M3 and lifetime-safe adaptive mixed backward specializations both remain at 128 registers without spills.

From `TK_quantisation/mxftp4_v3/`, the upstream dO extension was also rebuilt with:

```
make -B -j1
```

Both exact single-job builds exited with status zero.

### C.3 GPU discipline and validation

All final measurements use visible GPU 0. Before a GPU command, the device is checked independently:

```
nvidia-smi --id=0 \
  --query-gpu=utilization.gpu,memory.used \
  --format=csv,noheader,nounits
```

Timing runs are sequential. Representative validation entry points are:

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
  python3 validate_bf16_dq_path.py
```

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
  python3 validate_projection_fp4_epilogue.py \
  --sequence 4096 --heads 24 --hidden 3072
```

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
  python3 validate_unified_qkv_projection.py \
  --sequence 4096 --heads 24 --hidden 4096
```

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
  python3 validate_unified_qkv_projection_rope.py \
  --sequence 4096 --heads 24 --hidden 3072 \
  --warmups 4 --samples 13
```

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
  python3 validate_inverse_rope_nvfp4_handoff.py \
  --sequence 4096 --heads 24 --warmups 7 --samples 31
```

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
  python3 validate_upstream_mxftp4_dout.py \
  --sequence 8192 --heads 8
```

```
CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
  python3 validate_dq_projection_handoff.py \
```

```

—sequence 8192 —heads 8 —warmups 7 —samples 31 \
—repeat-checks 4

CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
python3 validate_projection_dgrad_path.py

CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
python3 evaluate_compact_dq_projection.py \
—shapes 8192x8,4096x24 —seed 20260813 \
—warmups 7 —samples 31

CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
python3 evaluate_llama_attention_e2e.py —rope —scope both \
—shapes 4096x24x4096,8192x24x4096,4096x64x8192,\
8192x64x8192,16384x64x8192 \
—seed 2026081402 —warmups 4 —samples 13 \
—output ../.. / results/llama_dense_attention_e2e_rope_broad_20260813.json

CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
python3 evaluate_llama_attention_e2e.py —rope —scope both \
—shapes 4096x24x3072,8192x16x2048,8192x24x4096,\
4096x64x8192,8192x64x8192 \
—warmups 5 —samples 11 —peak-tflops 2500 \
—output ../.. / results/llama_dense_attention_e2e_projection_native_broad_mfu_20

CUDA_VISIBLE_DEVICES=0 PYTHONPATH=../.. \
python3 -m tk_fa4.lowp_fa4_bwd.evaluate_llama_attention_convergence \
—shape 4096x24x3072 —steps 500 —learning-rate 2e-5 \
—relative-noise 0.2 —eval-interval 50 —seed 2026081417

```

## C.4 Primary result artifacts

## C.5 Correctness gates

The retained result is accepted only when:

1. every output is finite and has the expected shape/dtype;
2. all four E2M1 layouts and projection-native MXFP4 V match their standalone publishers bit-for-bit, including publication from the fused rotated BF16 fragment;
3. pair-native projection weights plus fused RoPE agree with dense split-half RoPE to the stated BF16 rounding floor; inverse RoPE returns Q/K gradients to the projection basis and leaves dV bit-exact;
4. forward and backward compact Q/K views alias one allocation;
5. fused dO row/column payloads match the split publishers bit-for-bit and the delta agrees with its FP32 reference;

**Table 33:** Evidence files used by this report.

| Artifact                                                                    | Contents                                                                                                                                         |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| results/lowp_fa4_bwd_bf16dq_validation_20260812.json                        | six-shape BF16 dQ latency, quality, interface agreement, and repeat stability                                                                    |
| results/lowp_fa4_bwd_breakthrough_experiments_20260812.md                   | BF16 dQ reduction, ceilings, profile summary, and projection-backward chain                                                                      |
| results/lowp_fa4_bwd_projection_fusion_20260812.md                          | native NVFP4 learned projection, unified QKV publication, layout validation, dispatch, and timing                                                |
| results/unified_qkv_projection_rope_packedbf16_s4096h24k3072_20260813.json  | pair-native fused-RoPE accuracy, byte-layout checks, inverse-gradient roundtrip, and component timing                                            |
| results/llama_dense_attention_e2e_rope_broad_20260813.json                  | ten-row RoPE-aware dense decoder boundary timing and component accuracy                                                                          |
| results/llama_dense_attention_e2e_projection_native_broad_mfu_20260814.json | five-shape FP4-forward activation/training timing, useful FLOPs, MFU, and stacked-gradient accuracy                                              |
| results/llama_attention_convergence_s4096h24_3seed_summary_20260814.json    | aggregate of three 500-step full-weight teacher/student trajectories; the three per-seed checkpoint histories are stored in the referenced files |
| results/lowp_fa4_bwd_adaptive_per_head_20260812.md                          | robust scale rule, regimes, producer timing, and broad accuracy                                                                                  |
| results/lowp_fa4_bwd_adaptive_outlier_check_20260812.json                   | full-rank sparse-outlier check                                                                                                                   |
| results/lowp_fa4_bwd_1to4_broad_eval_20260812.md                            | fixed mixed-dP broad matrix and accuracy frontier                                                                                                |
| results/lowp_fa4_bwd_1to6_frontier_20260812.md                              | preprocessing, forward-emitted V, rejected schedules, and multi-regime study                                                                     |
| results/lowp_fa4_bwd_adaptive_mixed_screen_20260813.json                    | two-shape, two-seed calibrated/unit/outlier validation of lifetime-safe adaptive mixed dP                                                        |
| results/lowp_fa4_bwd_compact_dq_projection_20260813.json                    | BF16/FP8/NVFP4/MXFP4 projection quality, charged timing, and composed dQ chain                                                                   |
| results/lowp_fa4_bwd_nv_mx_format_screen_20260813.md                        | pure-backward NVFP4/MXFP4 dS and compact-Q/K combination screen                                                                                  |
| results/lowp_fa4_bwd_bf16dq_source_profile_20260812.ncu-rep                 | source-correlated direct-BF16 profile                                                                                                            |
| results/tk_fp4_bwd_v382_ladder_20260803.md                                  | initial full-FP4 represented-operand ladder                                                                                                      |

6. direct BF16 dQ agrees with the BF16 value underlying the FP32-return route to the non-deterministic reduction floor;
7. repeated launches do not corrupt phase state;
8. wider-head direct-projection requests select the exact BF16-dQ plus vendor-GEMM fallback unless the experimental topology is explicitly forced;
9. precomputed-scale NVFP4 publication is bit-exact with the full quantizer when both receive the same device-resident global scale;
10. compact NVFP4 projection leaves dK/dV bit-exact and is finite before timing; experimental pure-format gates are restored to zero;
11. the unsigned stacked route agrees with the selected low-precision dX to the stated quantization floor, while activation-only results are not presented as weight-gradient-capable training steps;
12. convergence summaries expose both raw native validation loss and the independently measured low-precision teacher floor;
13. the hot specialization has no local stack or spills;
14. timing is reported only after the corresponding correctness gate.

The 100-call S8192/H8 repeat test remained finite. dK and dV were repeat-exact; maximum dQ drift was  $9.54 \times 10^{-7}$ .

## D Operand and API Contract

### D.1 Adaptive bundle

**Table 34:** Fields in B300AdaptiveLowpOperands.

| Field       | Shape / dtype          | Consumer                              |
|-------------|------------------------|---------------------------------------|
| q_fp4       | $[B, H, 192, S]$ uint8 | aligned sequence-major Q for dK       |
| score_q_fp4 | $[B, H, S, 96]$ uint8  | compact Q for score/dK                |
| k_fp4       | $[B, S, H, 192]$ uint8 | aligned depth K for dQ                |
| score_k_fp4 | $[B, H, S, 96]$ uint8  | compact K for score/dQ                |
| qk_scales   | $[B, H, 7]$ float32    | packing, score page, dQ/dK correction |
| mixed_v     | optional opaque tensor | reusable fixed mixed-dP V row         |

### D.2 Unified QKV bundle

The typed Q/K tensors returned by `forward_operands()` are views of `backward.score_q_fp4` and `backward.score_k_fp4`. Their data pointers are identical; the API does not allocate or copy a second payload.

The aligned byte containers reserve 16 bytes for every 16 logical E2M1 values: eight packed payload bytes followed by eight zero bytes. Compact depth rows reserve 96 bytes for 192 values. The layout



**Table 35:** Fields in B300UnifiedLowpQKV.

| Field                  | Shape / dtype                    | Contract                                                        |
|------------------------|----------------------------------|-----------------------------------------------------------------|
| q, k                   | optional $[B, S, H, 192]$ BF16   | ordinary publication; absent when <code>store_bf16=False</code> |
| v                      | optional $[B, S, H, 128]$ BF16   | ordinary publication; absent when <code>store_bf16=False</code> |
| backward               | B300AdaptiveLowpOperands         | exact aligned and compact Q/K byte views                        |
| q_forward_scales       | $[B, S/128, 3H, 512]$ E4M3       | canonical local Q scale pages                                   |
| q_forward_global_scale | $[B, H]$ FP32                    | $1/(448m_h)$                                                    |
| k_forward_scales       | $[B, S/64, 3H, 512]$ E4M3        | canonical local K scale pages                                   |
| k_forward_global_scale | $[B, H]$ FP32                    | $1/(448m_h)$                                                    |
| v_forward_fp4          | $[B, H, 128, S/2]$ E2M1x2        | transposed MXFP4 V payload                                      |
| v_forward_scales       | $[B, H, 2, S/128, 32, 16]$ uint8 | V E8M0 scale pages                                              |

tensors are opaque to model code; consumers should pass the bundle rather than reinterpret the byte strides.

### D.3 Public producer levels

**Table 36:** Public adaptive Q/K producer interfaces.

| Interface                                   | Contract                                                                                                            |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| b300_mha_fwd_with_adaptive_lowp             | run attention forward while an auxiliary producer computes robust scales and packs Q/K                              |
| b300_adaptive_lowp_operands_from_scales     | pack BF16 Q/K with an already-available seven-word per-head record                                                  |
| b300_adaptive_lowp_operands_from_projection | zero-copy validation wrapper for four layouts emitted elsewhere                                                     |
| b300_project_qk_adaptive_lowp_nvfp4         | run native NVFP4 learned projection and automatically choose fused or separate publication                          |
| b300_project_qkv_unified_lowp_nvfp4         | run one native NVFP4 QKV projection; publish shared Q/K, MXFP4 V, all backward Q/K views, and optionally BF16 Q/K/V |

### D.4 Upstream dO bundle

The fused dO producer writes a row-major E2M1x2 tensor  $[B, H, S, 64]$ , a feature-major E2M1x2 tensor  $[B, H, 128, S/2]$ , their E8M0 scale pages, and FP32  $\delta$  with shape  $[B, H, S]$ . These tensors are an upstream ownership contract: backward consumes them without running another row quantizer, column quantizer, or dot-product kernel.

### D.5 Backward endpoints

b300\_mha\_bwd\_adaptive\_lowp consumes only the retained winner route. Its default returns FP32 dQ for compatibility. Setting `return_bf16_dq=True` returns projection-ready BF16 dQ and omits

the widening pass.

`b300_mha_bwd_adaptive_lowp_projection_dgrad` writes one interleaved BF16  $[dQ, dK, dV]$  buffer and computes Equation 73. Projection weights use PyTorch's `[out, in]` convention and are packed once in per-head  $[Q_{192}, K_{192}, V_{128}]$  order.

## D.6 Current semantic boundary

The legacy adaptive bundle still accompanies ordinary BF16 Q/K because saved forward values, API shape checks, and non-low-precision consumers use them. The unified projection can suppress BF16 Q/K/V, but doing so is legal only when every selected forward and backward consumer accepts the low-precision bundle. The retained FP4+FP8 backward and the experimental all-MX path do not yet form one universal no-BF16 endpoint. That API promotion should follow composed forward/backward validation rather than precede it.